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Strong-Lensing Time Delays in the Berger–Hopf Framework: An R1 Calculation and an Observed-Light Test

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Abstract

Prepared environmental light probes a state-conditioned optical transfer in the Berger–Hopf interpretation. We calculate a regular finite-body response from the specified R1 cubic scalar–tensor action, obtaining fixed-preparation delay offsets of 25.9593 s and 8.6490 s. A positive conventional effective mass reproduces the retained metric, limiting state identifiability. We construct an HE 0435-1223 observed-light baseline and test separately frozen predictions. Brightness undercoverage, conditional HST central-image tension and a 3.26 mas Gaia discrepancy motivate source and geometry revisions. A positive G1 perturber reduces consumed-Gaia coordinate RMS to about 0.011 mas without enlarging errors, but the new JWST test fails instrumental adequacy. Correcting native-rate eligibility restores 303 valid slopes and gives about 0.45 mas raw geometry RMS for both reference and R1-inclusive controls; all sixteen profiles still fail. Actual read-group calculations evaluate an electron-conserving temporal mean, while a coherent observed-sky calculation propagates shared calibration uncertainty. The temporal term worsens the held optical-profile prediction. A provisional coupled sky/response step lowers training residuals but fails a held target exposure. These development applications specify the remaining measurement and physical-state dependencies; they establish neither precision validation nor a BHSM detection or unique envelope-state measurement.

Keywords: Berger–Hopf framework; strong gravitational lensing; time delays; envelope response; Vainshtein screening; effective-mass representation

1. Introduction

In the Berger–Hopf Standard Model (BHSM) interpretation, gravitational lensing is the optical response of an envelope event traversed by environmental radiation. The BHSM program asks how envelope-state and surface contributions determine that response. Prepared incoming light is related to outgoing image directions and observer arrival-time differences, conditional on the envelope state and surrounding geometry. This raises a modeling question: which properties of a state-conditioned transfer can be identified once ordinary lens and source freedoms are admitted?

We address that question through one explicit conditional realization. The previously specified closed reference branch R1 supplies a cubic scalar–tensor action and frozen rolling coefficients [1]. Its regular finite-body source determines the metric and projected encounter without a fitted local amplitude. The same emitted marker, physical source coordinate, measured redshifts, observer clock and compatible distance maps [2] are used for the response and its conventional fixed-body reference. This common preparation makes their difference a defined forward calculation.

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Cubic kinetic braiding, screening and their effects on lensing are established mechanisms [3–5]. Lensing inference also depends on the assumed profile, mass sheet, source and propagation geometry [6–9]. Our contribution is a controlled finite-body forward benchmark together with an explicit test of what its image and delay response identifies. It combines the normalized scalar source, one consistent potential/gradient/Hessian, resolved image branches, a positive conventional effective-mass representation and a finite nonlinear nuisance refit. Smooth-profile and distributed-encounter controls test whether the conclusions depend on the convenient body or optical plane. This is the practical role of a bounded forward model with stated limitations [10].

The calculation follows state and preparation, normalized source and boundary problem, metric/optical transfer, images and delays, then identifiability and approximation controls. A positive effective mass reproduces the retained scalar metric and its complete optical response. The reference-relative seconds therefore quantify a conditional transfer; they do not by themselves identify a unique underlying state. The finite refit makes this distinction concrete without changing the injected preparation. We then construct an observed-lens timing and point-geometry baseline for HE 0435−1223 and test frozen conventional and conditional R1 predictions against reserved later light curves. This applies the same identifiability question to actual observations without promoting the synthetic profile to a complete galaxy reconstruction. Its coverage failure and extra central image motivate a distinct HST-constrained source revision, followed by a frozen test on unused LCO catalogue targets. This sequence tests the ordinary source and predictive uncertainty before any residual is assigned to an additional state or interface response. The shared LCO reduction also motivates a separately frozen Gaia astrometric test of the same physical predictions. Its failure leads to an explicit diagnosis of the circular-deflector/constant-tide geometry, followed by a positive nearby G1 perturber in the full two-plane equations. Gaia is consumed in that revision. A further, separately frozen native JWST experiment tests its predictions without fitting the new target geometry or enlarging the errors. Its failed calibration motivates a separately labeled development calculation on those consumed photons: corrected native-rate selection, an electron-conserving read-history mean and a coherent sky with shared calibration covariance. Their actual residuals distinguish a repaired implementation from an adequate precision measurement. This extends the chain from a conditional optical transfer to the measurement operator that would be needed to infer its state dependence.

BHSM determines the research question and interpretation. The R1 action is an effective representation with its own independent rolling scalar; a unique physical reduction of BHSM cap/interface data to that action requires additional constitutive and embedding data. Those source conditions are derived separately in Supplementary Note S4. The present paper establishes the forward response and its information limits at the level actually specified. Its profiles are controlled examples; the observed-lens application is a restricted retrospective test and does not claim an empirical BHSM detection.

2. Envelope State and Environmental Preparation

The incoming marker probes the event; it does not select the envelope reference state. The conditional state consists of the frozen R1 branch, the finite local density excess and the regular local boundary branch specified below. The earlier source-free target state is retained as a separate null control in Appendix A. Homogeneous geometry, local matter and the local scalar response are accounted for once. This section specifies the synthetic preparation. The distinct observed R-band channel and its measured redshifts are introduced in Section 8; its two hypotheses share that same observational preparation and variability likelihood.

2.1. Incident light, finite body and environment

The lens and source redshifts are $z_l = 0.3$ and $z_s = 1$. The incoming test light is a narrow transient marker with identical emission phase, spectrum, polarization and physical transverse source coordinate in the effective response and its reference. The observed reference wavelength is 550 nm. The fixed source displacement is $s = (0.0014576633674174603, 0)$ Mpc. The light intensity is taken in the test-field limit, so its gravitational backreaction is neglected. The observables are image directions and observer proper-time differences between copies of that marker.

The local body in the conditional envelope is prescribed as a finite excess density above the homogeneous environment. The effective response and fixed-body reference use this same profile:

$$\rho(r) = \frac{15M}{8\pi R^3}\left(1 - \frac{r^2}{R^2}\right) \quad (0 \le r \le R), \qquad M = 10^{11} M_\odot, \quad R = 3 \text{ kpc}. \tag{1}$$

It vanishes outside the body. This is the quadratic-density Tolman–VII profile [11], used as a controlled spherical mass distribution. Define

$$m_g = \frac{GM}{c^2}, \qquad f_M(r) = \frac{M(<r)}{M} = \begin{cases} \frac{1}{2}(5x^3 - 3x^5), & x = r/R \le 1, \\ 1, & r \ge R. \end{cases} \tag{2}$$

For the effective scalar response, the density is prescribed and the leading pressure adjustment is calculated separately. We do not reuse the exact Einstein-fluid pressure as an exact modified-gravity solution.

The physical Jacobi maps are independently initialized for observer–lens, observer–source and lens–source propagation. They share one background, the same transported screen basis, and frequency normalization. At the reference state their scalar values are

$$(D_{ol}, D_{os}, D_{ls}) = (945.162839373,\ 1689.312239687,\ 1076.994348060) \text{ Mpc}. \tag{3}$$

The main body is excluded from these maps and enters the local encounter once. The local response calculation holds the maps fixed, isolating the encounter contribution. Its fixed-body reference uses the same R1 geometry; it is not a fit of a separate complete ΛCDM cosmology.

3. Action, Normalization, Local Source and Boundary Data

3.1. Conditional BHSM state representation

The question is how an envelope state determines a light transfer, and which state information that transfer identifies. At the effective level used here, the state specifies the rolling values H, v, d, the cubic normalization λ, the finite matter source and the regular boundary branch. The normalized action below makes their source and metric applications explicit. This is the state-conditioned forward problem used throughout the paper.

A unique microscopic map from BHSM cap/interface variables into this independent scalar action is not assumed. The cap momentum and source pullback, including their normalization and complement dependence, are worked out in Supplementary Note S4. They specify additional physical selection data rather than another adjustable lensing contribution. The present light calculation measures properties of the conditional metric response. Its equivalence to a conventional effective mass is then a statement about what those observations can identify.

3.2. Effective action and frozen R1 normalization

We use signature $(-,+,+,+)$ and $M_{\rm Pl}^{-2} = 8\pi G$. In the action and local field equations, $c = \hbar = 1$. The conditional R1 representation uses the retained scalar–tensor action

$$S = \int \sqrt{-g}\,\mathrm{d}^4x \left[\frac{M_{\rm Pl}^2}{2} R - \frac{1}{2}(\nabla T)^2 + \frac{(\nabla T)^2 \Box T}{\lambda} - V_0 + \mu^3 T + \mathcal{L}_m \right], \qquad \lambda = \Lambda^3 > 0. \quad (4)$$

The field T is an independent scalar. It is distinct from the target field η and from a metric conformal coordinate. For example, $\gamma_{ij} = a^2 e^{2\psi_{\rm geo}} \bar{\gamma}_{ij}$ gives $\psi_{\rm geo} = -\Psi$ to linear order in Newtonian gauge; it does not supply another independent scalar in addition to the metric.

The dimensionless background convention is $\widehat{T} = T/M_{\rm Pl}$, $\widehat{v} = \dot{\widehat{T}}/(M_{\rm Pl}H_0)$, $\widehat{d} = \ddot{\widehat{T}}/(M_{\rm Pl}H_0^2)$, $\widehat{\lambda} = \lambda/(M_{\rm Pl}H_0^2)$. The frozen parameters are

$$\begin{aligned} (\Omega_{m0}, \Omega_{r0}, \Omega_{k0}) &= (0.31, 9\times10^{-5}, -0.018), \\ H_0 &= 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \widehat{\lambda} = 1, \qquad \frac{\mu^3}{M_{\rm Pl}H_0^2} = 3, \\ \frac{V_0}{M_{\rm Pl}^2 H_0^2} &= 2.3986073449708067. \end{aligned} \quad (5)$$

They define one reference branch, not a framework-wide calibration. The stored background solution is reused without refitting. R1 serves as a reproducible fiducial background for this transfer benchmark; no fit to current cosmological observations or claim of phenomenological viability is made here. The reference spatial curvature is $K = -\Omega_{k0}H_0^2/c^2 > 0$. Its unperturbed angular-diameter maps obey $D_{ab} = a_b \sin[\sqrt{K}(\chi_b - \chi_a)]/\sqrt{K}$. The previously computed six-amplitude sensitivity of one normalized axial $n = 2$ harmonic is retained as a separate propagation control; its amplitudes are zero in the reference maps used for the local comparison. Its homogeneous scalar current,

$$J = \dot{\bar{T}} + \frac{6H\dot{\bar{T}}^2}{\lambda}, \qquad \dot{J} + 3HJ = \mu^3, \quad (6)$$

also provides a sign and normalization check on the local variation.

3.3. Reduction from the retained effective action

Varying Equation (4) gives

$$\Box T + \frac{2}{\lambda}\left[(\nabla_\mu \nabla_\nu T)^2 - (\Box T)^2 + R_{\mu\nu}\nabla^\mu T \nabla^\nu T \right] - V_T = 0. \quad (7)$$

The Ricci term couples the scalar response to matter through the same Einstein equation. No independent phenomenological matter coupling is introduced. Write $T = \bar{T}(t) + \pi_T(r)$ and freeze $v = \dot{\bar{T}}$, $d = \ddot{\bar{T}}$ and H at the lens epoch. After subtracting the homogeneous equations, retain leading weak metric potentials and quasi-static derivatives, including the nonlinear spatial Hessian responsible for screening. Set

$$b_T = \frac{2v^2}{\lambda}, \qquad C = 1 + \frac{4(d + 2Hv)}{\lambda} - \frac{2v^4}{M_{\rm Pl}^2\lambda^2}. \quad (8)$$

The leading common constraints are

$$2M_{\rm Pl}^2 \Delta\Phi = \delta\rho - b_T \Delta\pi_T, \qquad \Phi = \Psi, \quad (9)$$

$$C\Delta\pi_T - \frac{2}{\lambda}\left[(\Delta\pi_T)^2 - (\partial_i\partial_j\pi_T)^2 \right] = -\frac{b_T}{2M_{\rm Pl}^2}\delta\rho. \quad (10)$$

The last term of C follows from eliminating the metric constraint. The equal potentials are a result at the retained order; they are not an exact assertion about a fully nonlinear space-time.

At $z_l = 0.3$ the frozen solution yields

$$\begin{aligned} H/H_0 &= 1.1836105772, & \widehat{v} &= 0.2377644264, \\ \widehat{d} &= 0.2484715891, & b_T/M_{\rm Pl} &= 0.1130638449, \\ C &= 4.2388585594. \end{aligned} \tag{11}$$

The physical b_T has mass dimension one. In numerical units $\widehat{\lambda} = 1$ is fixed; it is not a new local adjustable length.

3.4. Regular finite-body branch

For spherical symmetry the scalar equation integrates to

$$Cr^2\pi_T' - \frac{4}{\lambda} r(\pi_T')^2 = -b_T G M(<r). \tag{12}$$

The origin is regular and carries no additional scalar charge. The root continuously connected to the decaying weak-source solution is

$$\pi_T' = -\frac{2b_T G M(<r)}{r^2\left[C + \sqrt{C^2 + 16 b_T G M(<r)/(\lambda r^3)}\right]}. \tag{13}$$

It has $\pi_T' \propto r$ at the center and no interface delta-function source at R.

3.5. Executed source and material-boundary application

The source consumed by this reduction can be made explicit. Define an additional metric energy source ρ_Σ and independent scalar Euler source J_T by the pre-elimination rows $2M_{\rm Pl}^2\Delta\Phi = \rho + \rho_\Sigma - b_T\Delta\pi_T$ and $A\Delta\pi_T - 2[(\Delta\pi_T)^2 - |\,\mathrm{Hess}\,\pi_T|^2]/\lambda + b_T\Delta\Phi = J_T$, where $A = C + b_T^2/(2M_{\rm Pl}^2)$. The actual reduced source and spherical flux are

$$\begin{aligned} J_{\rm eff} &= J_T - \frac{b_T}{2M_{\rm Pl}^2}(\rho + \rho_\Sigma), \\ Cr^2\pi_T' - \frac{4}{\lambda} r(\pi_T')^2 &= \int_0^r s^2 J_{\rm eff}(s)\,\mathrm{d}s. \end{aligned} \tag{14}$$

For the declared action there is no extra layer term: $\rho_\Sigma = J_T = 0$, while the volume source remains nonzero. For a proper-radial thin layer with coefficients σ_Σ and $j_{T,\Sigma}$, its retained weak-metric jump would instead be

$$[Cr^2\pi_T' - 4r(\pi_T')^2/\lambda]_{R^-}^{R^+} = R^2\left(j_{T,\Sigma} - \frac{b_T\sigma_\Sigma}{2M_{\rm Pl}^2}\right). \tag{15}$$

These are source applications of the original coupled equations; a surface charge is not an independently adjustable lensing amplitude. An additional anisotropic interface stress would also consume the corresponding metric slip equation.

Using the frozen coefficients, the computed surface section is $\pi_T'(R)/(M_{\rm Pl}H_0) = -2.119825778845613 \times 10^{-4}$. In $M_{\rm Pl} = H_0 = c = 1$ units, with radius in c/H_0, an independent density integration gives $\int_0^R r^2 J_{\rm eff}\,\mathrm{d}r = -1.216417408185084 \times 10^{-13}$. The maximum relative nonlinear-flux residual at $r/R = 0.1, 0.5, 1, 2, 10$ is 4.15×10^{-16}; the original integrated metric Gauss residual is zero at printed precision. Independent interior quadratic and exterior rational roots give a relative material-boundary flux mismatch 6.23×10^{-16}.

The separate exact-GR material control has continuous same-oriented extrinsic curvature and no material layer. That material worldtube is distinct from the cosmological cap seam, and its zero layer does not remove the nonzero volume source. The receipt is `artifacts/strong_lensing_bhsm_integration_20261020/source_application.json`. Supplementary Note S4 derives the cap/source pullback and shows constructively why the leading normalization does not select a unique microscopic completion.

4. Metric and Optical Encounter Response

The common metric constraint gives

$$\Phi' = \Psi' = \frac{GM(<r)}{r^2} - \frac{b_T}{2M_{\rm Pl}^2}\pi_T', \qquad F(r) \equiv \frac{\delta\Phi'}{\Phi_N'} = \frac{2\epsilon_\infty}{1+\sqrt{1+r_V^3 f_M(r)/r^3}}, \tag{16}$$

where

$$\epsilon_\infty = \frac{b_T^2}{2M_{\rm Pl}^2 C}, \qquad r_V^3 = \frac{16 b_T GM}{\lambda C^2}. \tag{17}$$

The last expression uses $c = 1$. The numerical conversion is

$$r_V^3 = 16(b_T/M_{\rm Pl})m_g(c/H_0)^2/(\widehat{\lambda}C^2).$$

Deep in the exterior screened region $F(r)$ approaches $2\epsilon_\infty(r/r_V)^{3/2}$; at large radius it approaches ϵ_∞. The potential itself still decays. This is the usual cubic screening mechanism [4,5], evaluated here with the frozen normalization and a regular extended body.

The corresponding leading hydrostatic correction is

$$\delta p(r) = \int_r^R \rho(s)\,\delta\Phi'(s)\,\mathrm{d}s, \qquad \delta p(R) = 0. \tag{18}$$

It is positive and changes the central Newtonian pressure by 3.56959×10^{-6} fractionally. This supports the specified leading-order matter configuration; it does not construct an exact modified Tolman–Oppenheimer–Volkoff sequence or prove radial stability. Holding density fixed and changing its required pressure does not impose an independently fixed common material equation of state on both models.

4.1. Projection into the optical encounter

Restore c and interpret Φ as the dimensionless metric potential. For physical impact radius $b_{\rm imp}$ the additional deflection is

$$\delta\hat{\alpha}(b_{\rm imp}) = 4b_{\rm imp}\int_0^\infty \frac{\delta\Phi'(\sqrt{b_{\rm imp}^2+z^2})}{\sqrt{b_{\rm imp}^2+z^2}}\mathrm{d}z, \tag{19}$$

$$\delta\hat{\psi}(b_{\rm imp}) = \int_R^{b_{\rm imp}} \delta\hat{\alpha}(s)\,\mathrm{d}s. \tag{20}$$

The arbitrary zero of potential does not affect relative delays. We calculate the gradient and Hessian consistently from this same projected potential. A local force ratio is not substituted for a projected photon deflection ratio.

The leading stationary delay change between image i and reference image 0 is

$$\delta\Delta t_{i0} = -\frac{1+z_l}{c}\left[\delta\hat{\psi}(b_i) - \delta\hat{\psi}(b_0)\right]. \tag{21}$$

The finite result also resolves the image positions in Equation (23). The projection freezes the local coefficients during the encounter and extends the isolated local solution through the thin plane. Matching to the linear perturbation is assumed in an overlap region $r_V \ll r \ll c/H_l$; a global closed-cosmology solution has not been constructed.

4.2. Which state coefficients enter the transfer?

The state dependence is explicit in $b_T = 2v^2/\lambda$ and $C = 1 + 4(d + 2Hv)/\lambda - b_T^2/(2M_{\rm Pl}^2)$. At fixed body, λ and physical radius, define $S_r = \sqrt{1 + r_V^3 f_M(r)/r^3}$. Differentiating the original force gives

$$\frac{\partial \ln \delta\Phi'}{\partial \ln b_T} = \frac{3}{2} + \frac{1}{2S_r}, \qquad \frac{\partial \ln \delta\Phi'}{\partial \ln C} = -\frac{1}{S_r}. \tag{22}$$

Deep screening suppresses the local force sensitivity to C. Projection in Equation (19) takes positive weighted averages of these coefficients. For the reference image impacts, ordered positive, central and negative, the projected logarithmic derivatives are

$$\begin{aligned} \partial_{\ln b_T} \ln \delta\hat{\alpha} &= (1.525461, 1.516634, 1.522208), \\ \partial_{\ln C} \ln \delta\hat{\alpha} &= (-0.050923, -0.033268, -0.044417). \end{aligned}$$

The outer field therefore retains nonzero C sensitivity even though the impact lies well inside r_V.

These derivatives concern the extra deflection; derivatives divided by the total deflection carry its additional small fractional contribution. They quantify the conditional state-to-transfer relation without changing the frozen prediction. A particular physical state variation must also satisfy the action's background and boundary constraints and be composed with its H, v, d to b_T, C map. Independently varying the two reduced coefficients is not asserted to produce another globally matched R1 state. Section 6 addresses the separate question of identifying those coefficients from light when the ordinary mass distribution is allowed to change.

5. Image Directions and Observer Delays

5.1. Image directions and observer arrival-time differences

Let $x = D_{ol}\theta$, $\beta = D_{os}^{-1}s$ and $\hat{\alpha} = \nabla_x \hat{\psi}$, where $\hat{\psi}$ has length units. The common thin-lens equations are

$$\beta = \theta - D_{os}^{-1} D_{ls} \hat{\alpha}(D_{ol}\theta), \tag{23}$$

$$ct(\theta) = \frac{1}{2}(\theta - \beta)^T \mathsf{T} (\theta - \beta) - (1 + z_l)\hat{\psi}(D_{ol}\theta) + f(\beta),$$

$$\mathsf{T} = (1 + z_l) D_{ol}^T D_{ls}^{-1} D_{os}. \tag{24}$$

The maps must satisfy reciprocity so that T is symmetric [2]. A common $f(\beta)$ cancels in relative delays. Image positions are resolved again after changing the encounter.

For smooth geometric variations, fixing s implies $\delta\beta = -D_{os}^{-1}\delta D_{os}\beta$; fixed physical and fixed angular source prescriptions cannot be imposed simultaneously. At fixed measured redshift, the stationary first variation is

$$c\delta t = \frac{1}{2} q^T \delta\mathsf{T} q - q^T \mathsf{T} \delta\beta - (1 + z_l)\left[\hat{\alpha}^T \delta D_{ol}\theta + \delta\hat{\psi}|_x\right] + \delta f, \qquad q = \theta - \beta. \tag{25}$$

The image-position variation cancels at a stationary image. In the local R1 comparison $\delta D_{ab} = \delta\beta = 0$ and the nonzero intrinsic term is retained. Appendix B records the measured-redshift convention for comparisons that vary the maps.

5.2. Evaluated response and reference-relative offsets

The screened branch predicts $r_V = 212.0292$ kpc and $\epsilon_\infty = 0.001507886$, or a 0.1507886% asymptotic force enhancement. At $r = R = 3$ kpc the local force enhancement is 5.06706 parts per million (ppm), whereas the deflection enhancement at the same impact radius is 11.7467 ppm. The ray samples the outer, less screened field as well as the field near closest approach.

Table 1. Leading-order image positions for the effective response and its fixed-body reference. Image 0 is the positive outer image, image 1 the central image, and image 2 the negative outer image. Printed digits specify reproducible model values, not absolute astrometric accuracy.

Image	Reference (arcsec)	R1 response (arcsec)	Offset (μas)
0	+0.835476034	+0.835482157	+6.1233
1	−0.082161940	−0.082161152	+0.7881
2	−0.657495525	−0.657499874	−4.3486

Table 2. Relative delays for the same source marker, with image 0 as reference. The offsets are relative to the specified fixed-body reference, with the physical light preparation held fixed.

Observable	Fixed-body reference	R1 response	Offset
Δt_{10} (days)	21.716076566	21.716377020	25.9593 s
Δt_{20} (days)	14.271452387	14.271552491	8.6490 s
$\Delta t_{10}/\Delta t_{20}$	1.521644467	1.521654847	6.8212 ppm

Figure 1 compares the radial response and the two delay changes. The nonzero response follows from the rolling scalar already in Equation (4). It does not alter the earlier constant-target null result in Appendix A. An independent singular isothermal control, with velocity dispersion 220 km s^{-1} and the same reference source, retains images at +1.067883 and −0.711922 arcsec and a 17.094532-day delay. That analytic control validates the optical conventions; the finite body in Tables 1 and 2 is the local boundary problem used for the evaluated response.

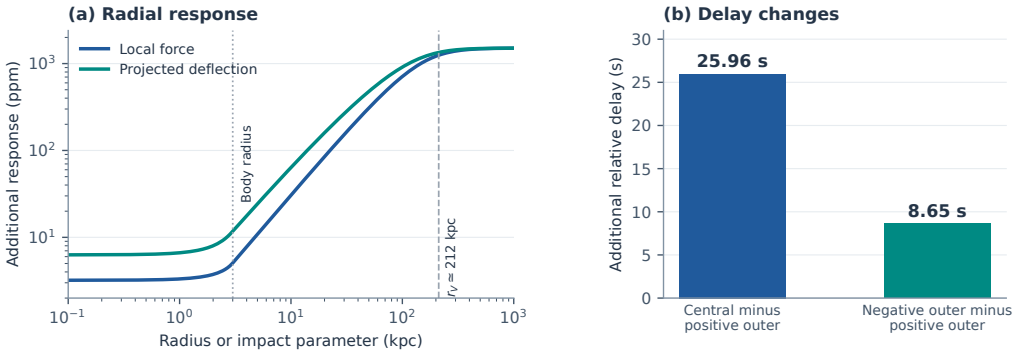


Figure 1. R1 envelope response relative to the fixed-body reference. Left: additional local force and projected deflection as fractions of their fixed-body reference values. The distinction reflects integration along the ray. Right: reference-relative offsets in the two delays for the fixed source preparation. Neither panel is an observational detection forecast.

At fixed epoch, $r_V \propto M^{1/3}$. The same frozen background also specifies the redshift dependence, without refitting:

Lens redshift	ϵ_∞	r_V (kpc), $M = 10^{11} M_\odot$
0.0	0.00351132	248.300
0.3	0.00150789	212.029
0.5	0.00086052	191.230
1.0	0.00022832	150.778
1.5	0.00007085	123.011

This is a prediction for the coefficients and force law, not a delay table for identical angular sources at these different redshifts.

The supplied reference-run outputs give the following implementation checks. The integrated scalar flux has maximum relative residual 7.9×10^{-16}. A direct line-of-sight quadrature and an independent angular-substitution quadrature agree on the additional deflection to 4.5×10^{-13} relative. The image equation residual is below 8.5×10^{-22} rad. Differentiating the projected potential and deflection independently gives relative discrepancies below 2.0×10^{-9} and 2.4×10^{-12}, respectively. Finite re-solved and stationary first-order delay changes differ by 0.00016 s and 0.000056 s. These are numerical and perturbative checks within the model, not bounds on omitted physics. In the current Windows replay, the corresponding flux residual is 8.30×10^{-16} and the deflection-derivative discrepancy is 3.19×10^{-12}; the remaining differences are recorded with their numerical tolerances in the replay receipt. The supplied and replay outputs remain separate records.

6. What the Light Transfer Identifies

6.1. Positive effective-mass representation of the same response

The retained metric and optical response admit an explicit conventional representation. Define an effective additional enclosed mass by

$$\delta M_{\rm eff}(r) = M f_M(r) F(r), \qquad \delta\rho_{\rm eff}(r) = \frac{M}{4\pi r^2}\frac{\mathrm{d}}{\mathrm{d}r}[f_M(r)F(r)]. \tag{26}$$

This density is nonnegative. To see this, put $S = \sqrt{1 + r_V^3 f_M / r^3}$ and $n_M = \mathrm{d}\ln f_M / \mathrm{d}\ln r \geq 0$. Then

$$\frac{\mathrm{d}\ln(f_M F)}{\mathrm{d}\ln r} = \frac{1}{2}\left[(1+S^{-1})n_M + 3(1-S^{-1})\right] \geq 0. \tag{27}$$

The ordinary Poisson field of $M(<r) + \delta M_{\rm eff}(r)$ has exactly the same $\Phi = \Psi$ as Equation (16) within the retained approximation. It therefore reproduces the full deflection field, relative potential, images and delays, including multiple source planes in the same geometry. Its asymptotic extra mass is

$$\delta M_{\rm eff}(\infty) = \epsilon_\infty M = 1.50788624 \times 10^8 M_\odot. \tag{28}$$

The rank-one multiple-source residual discussed below distinguishes the scalar from the restricted Tolman–VII parameter family, not from all conventional matter distributions. Because slow matter and light respond to the same retained potentials, ordinary dynamical data do not automatically break this equivalence either. Independent restrictions on the matter/environment model or an observable outside this retained equivalence would be needed to distinguish those representations. No such independent likelihood is asserted here, and such a distinction is not required for the envelope-transfer calculation.

The effective mass in Equation (26) represents the same retained contribution as the scalar correction in Equation (16). One uses either the explicit scalar correction or its

effective-mass representation in the encounter, not their sum. The construction proves equivalence of the retained metric and optical response. The physical pullback of microscopic BHSM surface variables into the normalized effective source is specified separately in Supplementary Note S4.

6.2. Single-source constraints and nuisance freedoms

For an observable residual $\delta y = y - y_0$, write

$$\delta y = B\,\delta a + N\,\delta \nu + \epsilon, \qquad WCW^T = I, \qquad P = I - (WN)(WN)^{+}. \tag{29}$$

Here B is a physical response, N the stated nuisance response, C a specified covariance, and $+$ a Moore–Penrose inverse. The rank of PWB concerns distinguishability in that nuisance model. A zero direction is unconstrained by the corresponding likelihood; a Fisher pseudoinverse does not turn it into a zero error bar. We take C positive definite. For the screened comparison, the single column is a fixed reference-relative response template; rank one denotes a surviving residual direction, not identification of a freely varied fundamental scalar parameter.

Using source position, body mass and radius, and a compensated mass-sheet scale as nuisances, the single R1-response column has projected rank one if the body center is fixed. Adding both center coordinates gives rank zero. The raw delay shifts in Table 2 are therefore not nuisance-independent observables for this configuration. The rank calculation uses 0.001 arcsec and 0.1 day as numerical whitening scales, not as demonstrated observational precisions.

The ordinary mass-sheet transformation,

$$\psi_\lambda(\theta) = \lambda\psi(\theta) + \frac{1-\lambda}{2}|\theta|^2, \qquad \beta_\lambda = \lambda\beta, \qquad \Delta t_\lambda = \lambda\Delta t, \tag{30}$$

cancels from the ratio of two delays. It does not remove source, centering, radial-profile, or independent line-of-sight freedoms [8]. Consequently the 6.8212 ppm ratio change is a fixed-preparation prediction, not a unique diagnostic of a distinct representation.

For the separately computed axial smooth-map response, the images of a softened isothermal model depend only on $\beta = s/D_{os}$, $\theta_E = b_{\rm phys} D_{ls}/D_{os}$, and $\theta_c = r_c/D_{ol}$, while all delays share the factor $(1+z_l)D_{ol}D_{os}/(cD_{ls})$. Varying the three scalar maps can therefore be reproduced by these source/profile changes and a delay scale. The saved six-anchor response accordingly has zero projected rank in the unrestricted nuisance model. This analytic factorization concerns that family; it is not a universal theorem that cosmological lensing information always vanishes.

6.3. A finite nonlinear conventional refit

A local rank calculation can be tested by a finite inverse application. We take the saved R1 images and delays as a deterministic injection and fit an ordinary Tolman–VII body with mass, radius, source position, lens center and one physical convergence sheet free. The injected marker, state, geometry and source remain the original preparation; fitted source coordinates are inverse-model nuisances. The fitted encounter contains the sheet once, while the propagation maps remain fixed. No independent delay scale is introduced.

For each of the three observed axial image positions, the original rows are $\theta_i - \beta - q\hat{\alpha}_{M,R}[D_{ol}(\theta_i - \theta_c)] - \kappa(\theta_i - \theta_c) = 0$, with $q = D_{ls}/D_{os}$. The two relative Fermat arrival rows use

$$\hat{\psi}(x) = \hat{\psi}_{M,R}(x - D_{ol}\theta_c) + \frac{\kappa}{2D_{ol}q}|x - D_{ol}\theta_c|^2. \tag{31}$$

All three image branches are re-solved after fitting. This is a bounded five-coordinate fit to five axial observables, a deliberately saturated local example whose Jacobian is nonsingular. It demonstrates finite absorption in this example; it is not a global uniqueness or detection claim. The free-center original lens and stationarity residuals are 4.24×10^{-22} rad and

Table 3. Finite conventional fits to the saved R1 injection. Entries are numerical mismatches after re-solving image branches, not observational errors or sensitivity forecasts.

Quantity	Center fixed	Center free
$\delta \ln M$	3.9960×10^{-6}	4.0206×10^{-6}
$\delta \ln R$	3.9827×10^{-7}	4.1234×10^{-7}
$\delta\beta$ (arcsec)	4.3495×10^{-9}	6.8148×10^{-9}
θ_c (arcsec)	0	1.7743×10^{-9}
κ	9.9469×10^{-6}	9.9229×10^{-6}
Maximum image mismatch (μas)	1.0143×10^{-4}	8.74×10^{-11}
First delay mismatch (s)	$+0.0033064$	$+9.21 \times 10^{-10}$
Second delay mismatch (s)	-0.0050312	-1.53×10^{-10}

1.30×10^{-18} Mpc. Allowing the ordinary model to adjust therefore removes the original 25.96 s and 8.65 s fixed-body offsets to numerical precision in this preparation. Fixing the center leaves a small finite residual; its significance would require an actual observing likelihood and independently justified lens priors. Supplementary Note S7 gives the original equations, parameter bounds, mass-sheet symmetry and independent-root checks.

6.4. A bounded multiple-source test

A supplemental design uses one body and three sources at $z_s = (0.8, 1.2, 1.8)$, with source offsets $0.1\theta_{E,s}$ at angles $(0.15, 1.1, 2.2)$ rad. These choices were fixed before applying the screened response. The nuisance model includes independent source positions and delay scales, a shared mass, radius and center, and shared physical convergence and two shear components. The resulting scalar column retains rank one; adding independent source-plane convergence columns leaves rank one as well.

This surviving component is small: its norm is 0.1202% of the raw whitened response in the expanded nuisance model. In the same illustrative weighting, the residual coordinates reach approximately 6 nanoarcsec and the residual delays 0.086 s. The explicit 24 observable weights and response arrays are supplied with the code. These tidal nuisance fields are phenomenological controls. This test does not solve the nonlinear scalar equation for a nonspherical galaxy and its environment, and it is not a signal-to-noise forecast. Its rank is specific to the finite radial profile family.

7. Robustness of the Transfer Calculation

7.1. A smooth profile with the same mass and half-mass radius

The body is a controlled input, so the response must be tested against its profile choice. We compare the compact Tolman–VII body with the standard smooth Plummer sphere [12], matching total three-dimensional mass and three-dimensional half-mass radius, rather than fitting a desired deflection. The reference half-mass radius is $r_h = 1.92941634649$ kpc. The Plummer scale is fixed by $a = r_h\sqrt{2^{2/3} - 1} = 1.47874508325$ kpc, with

$$\rho_P(r) = \frac{3Ma^2}{4\pi(r^2 + a^2)^{5/2}},$$
$$M_P(< r) = \frac{Mr^3}{(r^2 + a^2)^{3/2}}, \qquad \hat{\alpha}_P(b) = \frac{4m_g b}{b^2 + a^2}. \tag{32}$$

This profile has a regular center and finite total mass, with no hard material edge. Its consistent potential is $\hat{\psi}_P = 2m_g \ln[(b^2 + a^2)/(2a^2)]$. The same original scalar flux, regular root and metric constraint are applied to its enclosed mass. The background, clock and maps remain fixed. Both profiles have the same reference r_V because their total mass and R1 state are equal.

Eight paired applications use the two profiles and four settings: reference, total mass increased by 20%, half-mass radius increased by 20%, or physical source displacement increased by 20%. Only one ordinary input changes at a time, and the change is shared by the response and its same-profile reference. These are sensitivity controls, with no assigned probability or empirical uncertainty prior. All eight applications retain three images.

Table 4. Re-solved reference-relative delay offsets for two declared profiles. Each pair uses common preparation, state and ordinary inputs. The finite input changes are controls, not fitted uncertainties.

Shared control	Tolman–VII offsets (s)	Plummer offsets (s)
Reference	25.9593, 8.6490	22.4510, 9.6200
M increased 20%	32.6999, 9.8907	28.6743, 10.7843
r_h increased 20%	24.2697, 9.9458	20.2722, 10.1974
Source offset increased 20%	26.8452, 10.3945	23.5110, 11.5634

At matched M, r_h, changing the ordinary profile changes the two reference delays by approximately -167125 s and -124500 s, while changing the R1 offsets by -3.5083 s and $+0.9710$ s. The former are deliberate model changes, about 6438 times the largest reference R1 offset; they are not an estimated error on an observed lens. This shows why mass and one size alone do not fix the transfer. A ray samples the entire field, even though both profiles share the same scalar force at their half-mass radius.

The positive effective-mass construction applies to both profiles. Independent direct line-of-sight projection and re-solved images agree with their scalar representations within 5.56×10^{-17} arcsec and 6.14×10^{-10} s. The maximum original scalar-flux residual across the eight applications is 7.25×10^{-16} and the maximum image-equation residual is 2.55×10^{-21} rad. Both baseline profiles retain projected rank one with a fixed center and zero with a free center in the declared linearized nuisance family. Thus profile smoothness changes the quantitative response while preserving its retained conventional representation. Supplementary Note S6 gives the profile normalization, support and independent checks.

7.2. The ordinary finite-body control

For the conventional control, a regular Einstein-fluid interior can be matched to Schwarzschild. Its compactness is $m_g/R = 1.59514 \times 10^{-6}$. The leading optical application above uses the weak-field projection of this density.

7.3. A common second-order ordinary-gravity control

To test the effect of the first omitted conventional deflection, we expand the exact static GR optical scattering integral of the same Tolman–VII body through second post-Minkowskian order (2PM). Put $\varepsilon_g = m_g/R$, $y = b_{\rm imp}/R$ and $t = \sqrt{1-y^2}$. The additional radial deflection is

$$\delta\hat{\alpha}_{\rm GR2}(b_{\rm imp}) = 2\varepsilon_g^2 I(y), \tag{33}$$

$$I(y) = \begin{cases} \dfrac{15(\arcsin y - yt)}{4y^2} + y\left(\dfrac{15}{2}t - 4t^5 - \dfrac{48}{7}t^7\right), & 0 < y < 1, \\ \dfrac{15\pi}{8y^2}, & y \geq 1. \end{cases} \tag{34}$$

At the center $I(0) = 0$ and $I'(0) = -6/7$; the interior coefficient includes the leading fluid pressure and the conversion from areal radius to invariant asymptotic impact. At and outside the surface it matches the standard $15\pi m_g^2/(4b_{\rm imp}^2)$ exterior deflection [13]. The consistent plane potential is

$$\delta\hat{\psi}_{\rm GR2}(b_{\rm imp}) = 2\varepsilon_g^2 R J(y), \qquad J'(y) = I(y), \qquad J(1) = 0. \tag{35}$$

The supplied implementation evaluates I, I' and J analytically.

The R1 response and fixed-body reference receive the identical GR2 potential, gradient and Hessian, and both image systems are re-solved. The R1 coefficients and leading scalar profile are held fixed. The scattering impact is identified with the physical thin-plane impact in this control. This checks robustness to the common GR term; it does not compute the missing scalar–GR higher-order terms or a complete finite-distance 2PM cosmological arrival functional. The replay command is

```
python scripts/run_strong_lensing_gr2_control.py
```

7.4. Result of the common GR correction

At the original two exterior-image impacts, the GR2 term is 3.68 and 4.68 ppm of the leading conventional deflection. Its inclusion changes the conventional image coordinates by $(+1.35456, -0.01519, -1.72123)$ microarcsec and the conventional relative delays by $(+8.28473, +2.55245)$ s. The absolute correction is therefore material at the scale of the calculated reference-relative image shifts.

The response-minus-fixed-body-reference offsets are considerably more stable under this common-term check.

Table 5. Robustness of the reference-relative R1 offsets under the same GR2 correction in both calculations. The final column is the change in the reference-relative offset, not the absolute GR correction.

Response offset	Leading calculation	Common GR2 added	Change
$\delta\Delta t_{10}$ (s)	25.95925815	25.95932714	$+68.99$ μs
$\delta\Delta t_{20}$ (s)	8.64901938	8.64900924	-10.14 μs
$\delta\theta_0$ (μas)	$+6.12327625$	$+6.12326934$	-6.91×10^{-6} μas
$\delta\theta_1$ (μas)	$+0.78810716$	$+0.78810711$	-5.58×10^{-8} μas
$\delta\theta_2$ (μas)	-4.34859153	-4.34858643	$+5.09 \times 10^{-6}$ μas

The two relative changes in the reference-relative delay offsets are $+2.66$ and -1.17 ppm. The delay-ratio response remains 6.8212 ppm. Thus the nonzero reference-relative response survives this specific common-term control without changing its quoted precision in seconds or microarcseconds.

An independent mixed stationary-action derivative reproduces the change in the reference-relative delay offsets within 0.7 ns; this checks the small drift without subtracting four re-solved delays.

The analytic correction was also compared with direct null-geodesic scattering in the solved static GR fluid and exterior metric. At the three reference impacts, the exact deflection minus the 1PM and 2PM terms has magnitude below 1.8×10^{-16} rad for this body's compactness $\varepsilon_g = 1.59514 \times 10^{-6}$. A compactness extrapolation independently recovers the interior 2PM coefficient, and direct integration verifies $J' = I$. The latter checks establish the implemented ordinary-GR scattering coefficient, not a physical error bound for the scalar model or its cosmological embedding. The full corrected coordinates, delays and checks are supplied in `gr2_control.json` and the review outputs.

7.5. A distributed encounter with the same physical endpoints

The thin plane can be checked by distributing the same retained field along the ray. Use lens-scaled conformal coordinates $z = a_l(\chi - \chi_l)$, $x = a_l x_{\rm com}$ and $\kappa_K = K/a_l^2$, where $a_l = (1+z_l)^{-1}$. The observer and source lengths are $\ell_o = a_l\chi_l$ and $\ell_s = a_l(\chi_s - \chi_l)$; the fixed source endpoint is $X_s = (a_l/a_s)s$. These are determined by the original closed reference and its physical distance maps. Define $S_\kappa(d) = \sin(\sqrt{\kappa_K}d)/\sqrt{\kappa_K}$, $C_\kappa(d) = \cos(\sqrt{\kappa_K}d)$ and the background generating function

$$G_d(x_1, x_2) = \frac{C_\kappa(d)(x_1^2 + x_2^2) - 2x_1x_2}{2S_\kappa(d)}. \tag{36}$$

For a common half-window L, the retained paraxial arrival action is

$$ct[x] = \frac{1}{a_l}\Bigg\{ G_{\ell_o - L}[0, x(-L)] + G_{\ell_s - L}[x(L), X_s] + \int_{-L}^{L}\left[\frac{x'^2 - \kappa_K x^2}{2} - 2\Phi(\sqrt{x^2+z^2})\right]\mathrm{d}z\Bigg\}. \tag{37}$$

Its original ray equation and endpoint rows are

$$\begin{aligned} x'' + \kappa_K x &= -2\partial_x\Phi, \\ x'(-L) &= \frac{C_\kappa(\ell_o - L)}{S_\kappa(\ell_o - L)}x(-L), \\ x'(L) &= -\frac{C_\kappa(\ell_s - L)}{S_\kappa(\ell_s - L)}x(L) + \frac{X_s}{S_\kappa(\ell_s - L)}. \end{aligned} \tag{38}$$

The individual exterior wavefront curvatures are fixed by those physical endpoint data. At the plane their sum is $D_{os}/(D_{ol}D_{ls})$ and the source term is s/D_{ls}, so they recover the original thin-plane preparation. They are not a freely chosen split of that sum. The factor $1/a_l = 1 + z_l$ retains the same observer clock.

Both the fixed-body and R1 fields are solved on each window, using all three noncritical ray branches. Their distributed actions are compared with the thin-plane/Born projection of the same windowed fields. At $L = 20$ Mpc, the same-window thin-plane offsets are

Table 6. Distributed-path change of the reference-relative delay offsets compared with the same-window thin plane, evaluated by direct arrival subtraction. This is a control of the retained frozen weak-field optical action.

Half-window L (Mpc)	Change of $\delta\Delta t_{10}$ (ms)	Change of $\delta\Delta t_{20}$ (μs)
0.21203	−0.46019	+1.5548
0.84812	−0.57857	+1.1474
4.24058	−0.61238	+1.0321
20.00000	−0.61905	+1.0056

25.95922277 s and 8.64900576 s; the distributed offsets are 25.95860372 s and 8.64900676 s. Tighter collocation and potential quadrature change the latter offsets by less than 0.3 ns. These are refinement figures for that subtraction. A separate on-shell derivative integrated along a numerical interpolation between the two fixed optical fields gives the preferred finite-response changes −0.61904348 ms and +1.00640532 microseconds. They agree with the raw subtraction within 7 ns; the interpolation is a numerical integration device, not a physical state variation. These are numerical controls within this defined optical problem.

An analytic exterior-tail bound follows from $0 \le \delta\Phi' \le \epsilon_\infty m_g/r^2$. For two fixed Born impacts b_i, b_0, it gives

$$|\delta\Delta t_{i0}^{\rm tail}| \le \frac{1+z_l}{c}\,\frac{m_g\epsilon_\infty|b_i^2-b_0^2|}{L^2}. \tag{39}$$

The sharper integrated bounds at 20 Mpc are 35.04 and 13.47 microseconds for the two extra-field fixed-impact delays. They do not bound shifts caused by re-solving all images, an arbitrary environment, or a changed global state. The new control isolates finite thickness and path response with fixed background, source and clock. It does not add the scalar-dependent field-equation corrections needed for a full higher-order metric prediction. The complete boundary construction, force/action residuals, tail derivation and convergence data are in Supplementary Note S8.

8. An Observed-Lens Baseline and Withheld-Light Test

8.1. Data, calibration and reserved observations

We next apply the transfer-to-identifiability question to HE 0435−1223, selected for its resolved four-image geometry and public long-term COSMOGRAIL monitoring [14,15]. The adopted redshifts are $z_l = 0.4546$ and $z_s = 1.693$. HST point-image and galaxy astrometry comes from Courbin et al. [16], with the required additional 2 mas systematic uncertainty. The unavailable full positional covariance is approximated by a diagonal covariance for the published relative positions; we do not fit Einstein-ring pixels or published cosmological posteriors.

The endpoint of the earlier 884-epoch release fixes the chronological boundary at native day 57447.03918. The 2020 release reprocesses some historical photometry, so we use its single consistent reduction: 902 training epochs and 100 later epochs, each containing four R-band magnitudes and their errors. The historical release is not concatenated with the newer reduction. The catalogue and release page differ by half a day in their stated absolute epoch origin; native day differences suffice for all splits and delays here. This is retrospective chronological validation in a common reduction, not a prospectively blinded observing campaign.

The protocol was fixed before fitting. Variability, microlensing and macro parameters were fitted on historical observations only, and the prediction artifact was committed before target scoring. The primary test conditions on all 100 later image-B observations as an explicitly allowed driver and predicts the 300 reserved A/C/D magnitudes. It is a batch cross-image conditional prediction; some driver observations are later than a target date. A separate all-channel forecast reserves all 400 later magnitudes and conditions on training data only. Neither test refits parameters, photometric offsets or extrinsic curves against its targets.

8.2. The conditional physical and brightness applications

We retain the same fixed R1 reference calibration, recomputing its FLRW distance maps and local scalar coefficients at the observed redshifts. This common geometry defines the conventional galaxy-potential benchmark and the conditional R1 comparison; it is not a cosmological viability fit. The local body is a positive Plummer source with fitted mass and scale. A rear quadratic optical screen at the fixed conditional redshift $z_p = 0.8$ supplies asymmetric propagation. It is total optical boundary information, rather than an identified physical perturber or a solved same-domain scalar environment.

For physical screen potential $\widehat{\psi}_p(x_p) = x_p^T U x_p/2$ with symmetric U, the original two-plane ray equations are

$$\begin{aligned} x_p &= D_p\theta - D_{lp}\widehat{\alpha}_l(D_l\theta), \\ x_s &= D_s\theta - D_{ls}\widehat{\alpha}_l(D_l\theta) - D_{ps}Ux_p. \end{aligned} \tag{40}$$

Their elimination gives the compatible maps

$$\mathsf{D}_{ol} = D_l I, \qquad \mathsf{D}_{os} = D_s I - D_{ps}D_p U, \qquad \mathsf{D}_{ls} = D_{ls} I - D_{ps}D_{lp}U. \tag{41}$$

The same eliminated two-plane arrival action gives the time-delay matrix and observer-clock normalization used in Equation (24). No second sheet is added to the local body. Supplementary Note S9 gives the full action and Schur reduction, including the required pre-conjugate map conditions and the isolated spherical scalar boundary prescription.

The local field is still the original regular sourced solution. At the refitted R1 image-A impact $r = 7.74161173$ kpc, the executed section has $\pi_T'/(M_{\rm Pl}H_0) = -1.83519726 \times 10^{-4}$ and consumed source $J_{\rm eff}/(M_{\rm Pl}H_0^2) = -40596.7951$. In $H_0 = M_{\rm Pl} = 1$ units the integrated scalar left and source rows both give $-2.36867868266 \times 10^{-13}$; the corresponding metric Gauss rows both give $5.16669407301 \times 10^{-12}$. The maximum scalar-flux residual is 1.10×10^{-15} and the four metric-Gauss section residuals are below 3.80×10^{-16} relative. These numerical field residuals do not establish that the assumed galaxy profile is observationally adequate.

For observed magnitude m_{ik} at observer time t_{ik}, we apply the classical variability likelihood

$$m_{ik} = m_0 + b_i + q(t_{ik} - d_i) + e_i(t_{ik}) + \epsilon_{ik}. \tag{42}$$

Here q is a common stationary Ornstein–Uhlenbeck process, a standard empirical quasar-variability model [17]; e_i are independent slower processes for extrinsic variability, and b_i have proper broad Gaussian offset priors. Their covariances are $\sigma_q^2 e^{-|\Delta t|/\tau_q}$ and $\sigma_e^2 e^{-|\Delta t|/\tau_e}$, respectively. Quoted photometric variance plus fitted common jitter supplies the independent measurement noise approximation; the total brightness covariance remains correlated. This empirical likelihood does not replace a BHSM quantum-state kernel.

An exact nine-state Kalman filter evaluates the normalized likelihood. At source time $u = t - d_i$, its state is $(q, e_A, e_B, e_C, e_D, b_A, b_B, b_C, b_D)$, with OU transitions for the first five components and constant offsets. Smoothing over each permitted conditioning set gives predictive means and marginal variances. The joint test score is

$$\log p(y_{\rm target} \mid y_{\rm condition}, \widehat{\vartheta}) = \log p(y_{\rm condition}, y_{\rm target} \mid \widehat{\vartheta}) - \log p(y_{\rm condition} \mid \widehat{\vartheta}), \tag{43}$$

with all parameters $\widehat{\vartheta}$ frozen. It is not the sum of univariate marginal scores. Supplementary Note S10 derives these applications and compares them with independent dense-Gaussian formulas.

The empirical training delays relative to A are

$$(d_B, d_C, d_D) = (6.41984, -0.97473, 12.48799) \text{ days},$$

where positive means later arrival. Four fixed starts retain multiple local modes; two reach nearly the same best likelihood. Both fitted variability timescales hit their preset upper bounds, 1500 and 10000 days. Delay-curvature estimates depend materially on step size and are not calibrated confidence limits. The macro fits use the exact historical brightness likelihood after initialization. Their shared variability hyperparameters remain fixed: this

is a two-stage conditional empirical Bayes calculation, rather than a fully marginalized physical and stochastic posterior.

8.3. Frozen predictions and observed performance

The conventional macro fit predicts delays $(6.020014, 0.672696, 13.611423)$ days; the separately refitted R1 model predicts $(6.022659, 0.672780, 13.616498)$ days. Their astrometric residual sums are 12.714 and 12.695 over eight relative-coordinate rows. The conventional fit has $M = 10^{11.61394} M_\odot$, Plummer scale $a = 10^{0.70626}$ kpc and screen convergence 0.3844 within the declared bounds. This large conditional propagation nuisance is not an independently measured line-of-sight convergence.

Activating R1 at exactly the conventional fitted body, source and screen changes the three delays by $(19.2903, 0.9386, 29.0367)$ s. Separately refitted differences, approximately $(228.51, 7.23, 438.44)$ s, also contain nuisance reallocation and optimizer termination effects; they are not the fixed-body physical signal. The independent full two-plane and compact actions agree to 2.4×10^{-14} day. A positive conventional effective mass still reproduces the entire retained R1 transfer, and therefore its test score. That shared score is a structural equality, not an independent empirical confirmation.

Table 7. Frozen later-B-conditioned predictions of 300 withheld A/C/D magnitudes. Log scores are joint continuous predictive densities in the same magnitude units. The zero-delay row keeps the same frozen variability parameters; it is a control, not a separately optimized alternative model. Intervals condition on point-fitted parameters.

Transfer	Joint log score	RMSE (mag)	95% coverage
Free training delays	795.1985	0.101520	85.00%
Conventional macro	791.6238	0.101333	85.33%
Conditional R1 macro	791.6288	0.101333	85.33%
Zero-delay control	755.9755	0.101436	86.67%

The conventional/R1 log-score difference is only 0.00498, while the conventional-minus-control difference is 35.6483 under this frozen likelihood. These are conditional predictive comparisons, not Bayes factors or evidence for a uniquely identified BHSM state. The empirical free-delay baseline scores 3.5746 higher than the restricted conventional macro model. Figure 2 shows that the conventional and R1 curves overlap at plotted precision.

Observed interval coverage is below nominal: conventional image-A coverage is 62%, versus 99% for C and 95% for D. The shared free-delay baseline also undercovers A. This is a limitation of the brightness/uncertainty baseline, rather than a BHSM-specific residual. The all-channel forecast has RMSE 0.194945 mag and 75.5% coverage; the R1-minus-conventional log-score difference is again only 0.00501. These two factorizations use the same later observations and are not independent replications. Temporally correlated targets do not justify an independent-binomial significance calculation for these coverage fractions. This frozen experiment is not retuned after exposure. Section 9 uses these consumed observations for development and tests a separate revision on unused LCO exposures.

A broad stationary-image search, with original-projection refinement, finds five images in both models, including a central image with macro flux approximately 6.32% of A before microlensing, extinction and galaxy-light contamination. No measured central-image limit enters the point-astrometry fit. Consequently this application is a restricted timing/point-geometry baseline, not a complete observed galaxy or Einstein-ring reconstruction. The central-pixel test in Section 9 now evaluates that additional prediction; a complete Einstein-

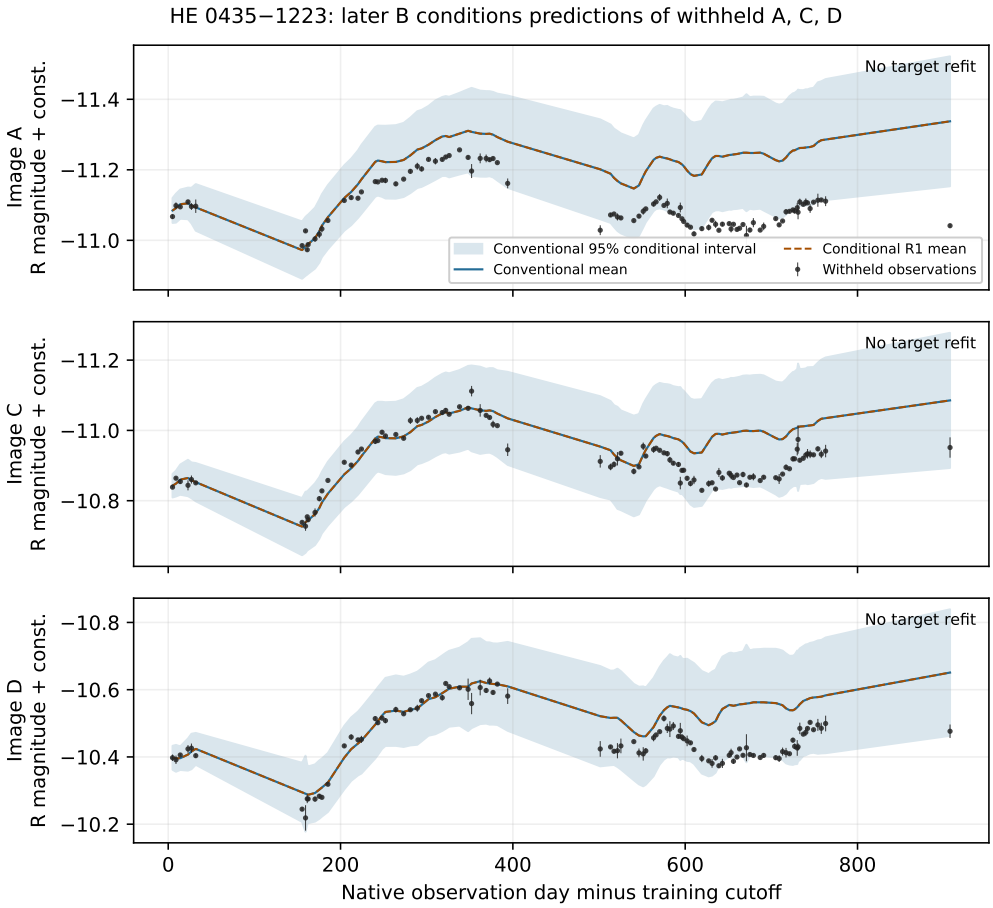


Figure 2. Reserved HE 0435−1223 A/C/D magnitudes and predictions conditioned on historical data and the allowed later-B driver. Lines join frozen predictive means at observation times; shading is the conventional 95% point-parameter conditional interval. No target fit or per-image magnitude correction is applied. Photometry: COSMOGRAIL/CDS [15].

ring reconstruction remains outside this family. Neither its existence nor the held-out brightness residuals are silently absorbed into a new envelope response.

9. HST Central Light, Revised Uncertainty and Unused Catalogue Targets

The first experiment is retained as an exposed development result. We now address its two concrete failures: overly restrictive brightness uncertainty and an additional central-image prediction that had not been tested against pixels. HST data constrain the revised source; a separate Las Cumbres Observatory (LCO) exposure set tests its subsequent predictions. Neither reuse of the first targets nor the HST-conditioned profile is counted as a new independent validation.

9.1. The original central-image prediction in HST pixels

We apply the original conventional 0.06320 and conditional R1 0.06313 central/A macro-flux predictions to public HST ACS/WFC F814W data (proposal 9744, 1445 s, 2003 August 18) and WFC3/IR F160W data (proposal 12889, 9939.59 s, 2012 October 28). Exact science, weight and context cutouts and full-product provenance are retained. Three field-star PSFs and an image-A empirical PSF probe template dependence. The four outer quasar images determine translation and rotation; the central template also receives independent freedom of ± 50 mas in each coordinate. The weight maps in these products are exposure-like, rather than inverse variance; drizzle correlations require separate attention [19]. Blank-sky scatter, relative exposure and source Poisson noise define the adopted marginal variance. The nominal bound also permits a 5% lower image-A flux normalization.

A deliberately flexible bound allows arbitrary nonnegative galaxy, host and unresolved nuclear light at every pixel. With $d_p = fP_p + B_p + \epsilon_p$, $P_p \geq 0$ and $B_p \geq 0$, a fixed-template Gaussian marginal-error union bound gives

$$f \leq \min_{p \in \mathcal{S}} \frac{d_p + \Phi_{\mathrm{N}}^{-1}(1 - 0.05/|\mathcal{S}|)\sigma_p}{P_p}, \qquad \mathcal{S} = \{p : P_p > 0.03 \max P\}. \tag{44}$$

Here Φ_{N} is the standard-normal CDF. Pixel independence is unnecessary. The 95% interpretation is conditional on the adopted PSF, sky subtraction and marginal-error model. Maximization over the finite PSF set and continuous centroid freedom is numerical, not a certified global or arbitrary-PSF coverage statement.

The F814W numerical envelope is 0.02935 of A, conservatively rounded up to 0.030. Multiplying every marginal noise scale by five and lowering the A normalization by 20% increases it to 0.04124, conservatively rounded to 0.042. Both are below the original 0.06320 prediction. The noise/normalization stress test is a sensitivity control, not an independent calibration of 95% coverage. F160W gives 0.554 and is inconclusive. Positive smooth galaxy profiles, including a free compact cusp and nuclear point, give the same direction of tension; their residual sums are descriptive, not detection significances. Figure 3 shows the pixel application. This tests the achromatic, unattenuated macro ratio. Differential extinction, microlensing or intrinsic variability across the central-image delay are not inferred here; retaining the old profile with such effects requires a separate physical flux prescription.

9.2. A revised source and classical brightness uncertainty

We retain a regular positive local source by adding a compact Plummer component to the outer body:

$$\rho(r) = \sum_{j=\mathrm{o,n}} \frac{3M_j a_j^2}{4\pi(r^2 + a_j^2)^{5/2}}, \qquad M(<r) = \sum_{j=\mathrm{o,n}} \frac{M_j r^3}{(r^2 + a_j^2)^{3/2}}. \tag{45}$$

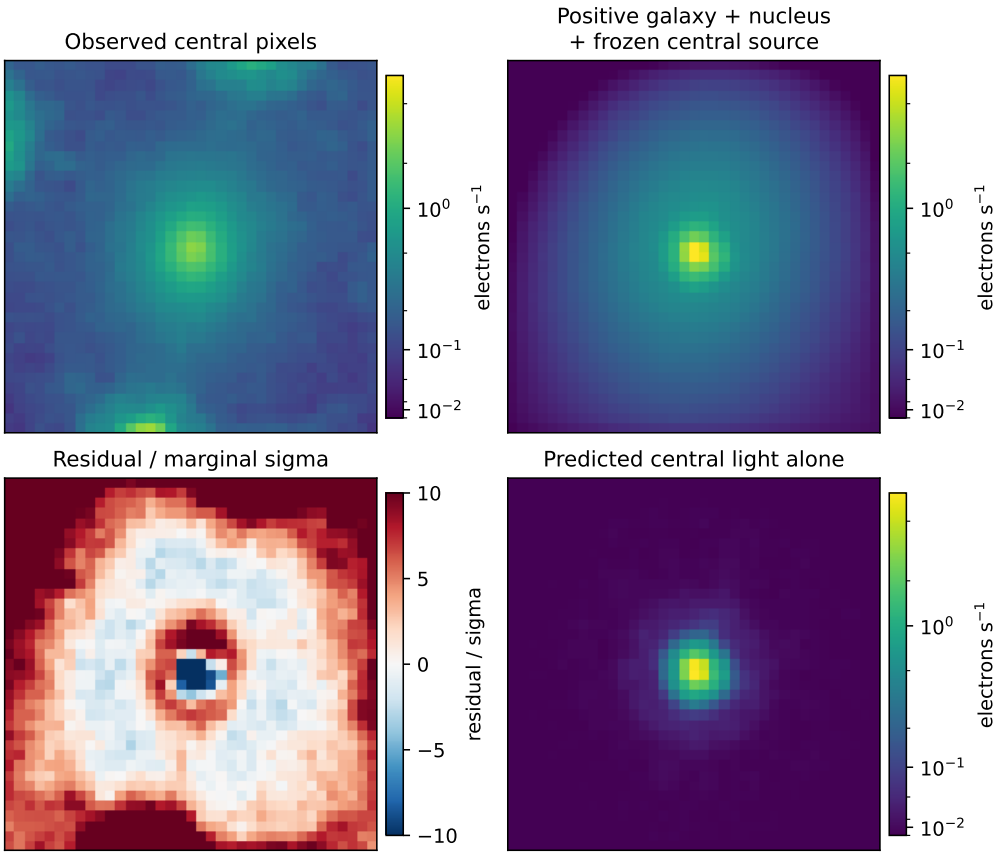


Figure 3. Actual ACS F814W pixels and the original central-image application. Upper left: measured central pixels; upper right: a fitted positive galaxy/cusp/nuclear-light model plus the frozen central source. Lower left: residual divided by the adopted marginal noise (colour range clipped at ± 10); lower right: predicted central light alone. Intensity colour bars are in electrons per second. The nonnegative-pixel test permits arbitrary positive contaminating galaxy/host/nuclear light. Its F814W conditional tension motivates the source revision. The separate F160W application does not exclude this prediction. HST data are development information in the revised model.

The compact scale is prescribed as $a_{\mathrm{n}} = 0.03$ kpc and its candidate masses are $0, 10^8, 10^{8.5}, 10^9, 10^{9.5} M_\odot$. This is a regular unresolved matter surrogate, not a black-hole mass measurement. The original R1 nonlinear equation consumes the *total* $M(< r)$ once; independently solved component scalars cannot be added. The original regular branch, metric Gauss equation, optical projection, rear screen and nine nuisance parameters are otherwise retained at this compact-source stage. The subsequent geometry revision is developed in Section 9.6.

For each conventional candidate, original astrometry and the exact development brightness likelihood determine the optical parameters. HST assesses the *sum* of every additional-image PSF, including unresolved multiple nuclear images. Among HST-admissible candidates the development objective selects one common compact mass for both models. The R1 model then refits its nuisance parameters using the same data. Candidate traces and the separate fixed-body R1 activation remain visible; this discrete profile selection does not define a posterior mass interval.

The revised classical brightness law is

$$u = t + \delta_k - d_i,$$
$$m_{ik}(t) = m_0 + b_{ik} + q(u) + e_i(u) + a_i \frac{u - t_0}{Y} + w_i(u) + \epsilon_{ik}, \qquad Y = 365.25 \text{ day}. \tag{46}$$

The source q and individual extrinsic processes e_i are independent stationary OUs. Gaussian slopes have standard deviation 0.03 mag yr^{-1}, and independent Brownian increments have covariance $D_i^2 \min(u - t_0, v - t_0)/Y$. Proper per-image/instrument offsets have 30-mag prior width, accommodating the distinct relative and apparent magnitude zeros without a target-derived offset. Quoted marginal errors plus per-image jitter supply measurement variance. Slopes and Brownian increments propagate increasing uncertainty beyond the last observation; all latent processes are conditioned jointly by the original state-space likelihood. Brownian increments start at zero at $t_0 = 52832$ in native day units, the floor of the earliest development epoch minus 31 days; this precedes every allowed shifted observation. None is a BHSM statistical-state contraction.

All 1002 old epochs and 25 pre-2019 LCO R epochs are development data. Three 500-day and three 2000-day rolling-origin windows recalibrate per-image uncertainty. The longer windows were added before numerical target inspection because known timestamps extend about five years past calibration. Overlapping development windows are not independent replications. The unchanged higher empirical 97.5th-percentile rule gives primary A/B/C/D scale factors $(1, 3.61321, 1, 1)$ and all-image forecast factors $(1.61012, 1.90726, 1.87754, 1.92479)$. In particular, short-window calibration alone misses longer-horizon B undercoverage.

A frozen 72-component finite nuisance ensemble propagates retained parameter variation and catalogue epoch-origin offsets $(-0.5, 0, +0.5)$ day. It is constructed from two short correlated chains and three development-only epoch-origin reweightings, not 72 independent posterior draws. Between-chain $\widehat{R}$ reaches 5.19 for weak extrinsic/Brownian decomposition and about 2.16 for source amplitude/time scale. Posterior convergence is not established; this is a specified empirical predictive algorithm, not a complete marginalized physical posterior. Fixing macro delays transports the ensemble with development likelihood-ratio weights. Driver conditioning subsequently updates weights and latent states only. The per-image affine calibration includes its joint-density Jacobian. We therefore assess both coverage and the cost in interval width and proper score on the new observations.

9.3. Unused LCO catalogue targets

The separate CTIO 1-m LCO monitoring [18] supplies 54 R epochs from 2019 through 2024 beyond every consumed COSMOGRAIL epoch. Its earlier 25 R epochs calibrate the instrument; all 80 V epochs remain unscored. Before opening R target values we fixed the split, model, calibration and source hashes and committed their predictive artifact. The primary application allows the 54 C magnitudes as a batch driver and withholds the 162 A/B/D magnitudes. C was selected using consumed first-experiment behavior. The secondary forecast withholds all 216 magnitudes and overlaps the primary test.

These numerical catalogue targets were unused by our model fitting and selection, with explicit limitations. The published LCO reduction uses a common difference-imaging reference mostly assembled from 2022; the exact reference-exposure list and full photometric covariance are unavailable. Common offsets capture static calibration uncertainty, but a full reference-noise likelihood is not identified by the released marginal errors. Numerical target values were unused; photon-disjoint calibration and full reference-noise independence are not established. The experiment is a retrospective catalogue-withholding test, not an independently blinded reduction or prospective forecast. Source identification also exposed qualitative published microlensing findings; numerical catalogue targets and plots were not inspected before freezing. Complete analyst blinding is not claimed.

9.4. Revised source and frozen diagnostics on unused catalogue values

The development selection retains candidate 1 with a regular compact matter component of $1.000 \times 10^8\, M_\odot$ and scale 0.03 kpc. The conventional and conditional R1 applications have 5 and 5 verified stationary images, respectively; all additional images enter the summed central-light constraint. The nearly degenerate development objectives do not measure a nuclear mass. The selected positive component is a conditional matter surrogate, not an inferred black hole or a solved stellar equation of state.

The original scalar and metric equations are applied to the total enclosed matter once:

$$Cx^2 p - \frac{4}{\widehat{\lambda}} x p^2 = -\widehat{b}\, m(x), \qquad 2x^2 \frac{d\Phi}{dx} + \widehat{b} x^2 p = 2m(x).$$

Here $x = rH_0$, $m(x) = GM(<r)H_0$, $\widehat{b} = b_T/M_{\rm Pl}$, $p = \pi_T'/(M_{\rm Pl}H_0)$ and $c = 1$, with the previously fixed $\widehat{\lambda} = \lambda/(M_{\rm Pl}H_0^2)$. At $r = 0.03$ kpc, the computed section has $\widehat{\rho} = \rho/(M_{\rm Pl}^2 H_0^2)$ equal to 3.734×10^9, reduced source $\widehat{J}_{\rm eff} = J_{\rm eff}/(M_{\rm Pl}H_0^2) = -\widehat{b}\widehat{\rho}/2$ equal to -1.712×10^8, and p equal to -3.599×10^{-5}. The scalar integrated left side and source are -3.495×10^{-17} and -3.495×10^{-17}, while the metric Gauss left side and source are 7.623×10^{-16} and 7.623×10^{-16}. Across the evaluated sections the maximum relative scalar and Gauss residuals are 3.419×10^{-16} and 2.168×10^{-16}. Activating R1 on the fixed conventional body, source, maps and clock gives B/C/D-minus-A delay offsets $(20.60284, 1.00422, 31.21607)$ s and a maximum coordinate image displacement $2.075 \times 10^1\ \mu$as. Separately refitted model predictions below need not retain those fixed-reference offsets.

The compact-source model's LCO test opens 162 previously unused A/B/D measurements, conditional on 54 allowed C measurements from the same later LCO epochs. The no-driver secondary forecast opens all 216 measurements. No brightness, delay, offset, calibration or source parameter is refitted to these target values. These scores belong to this frozen geometry, preceding the explicit G1 revision below. Table 8 reports the normalized joint conditional density and marginal predictive diagnostics. The joint log density is a density in the declared magnitude units; epochs are correlated, and its value is not a chi-square significance or an independent-event count.

Table 8. Frozen compact-source models on unused LCO R A/B/D observations, with image C as the allowed driver. Coverage is the empirical fraction within marginal 95% mixture intervals; width and the proper 95% interval score are in magnitudes. Larger joint log density and smaller interval score are preferred.

Model	Joint log p	RMSE	95% coverage	Width	95% interval score
Empirical delay ensemble	292.467	0.0780	98.77	0.6566	0.6640
Conventional macro	291.071	0.0772	98.77	0.6699	0.6779
Conditional R1 macro	291.071	0.0772	98.77	0.6699	0.6779
Fixed-body R1 activation	291.071	0.0772	98.77	0.6699	0.6779
Zero-delay control	289.016	0.0795	98.77	0.6245	0.6334

Table 9. Image-resolved primary diagnostics. Each image contributes 54 unused observations. These are correlated marginal diagnostics, not 54 independent coverage trials.

	Conventional			Conditional R1		
Image	RMSE	Coverage (%)	Width	RMSE	Coverage (%)	Width
A	0.0489	96.30	0.3667	0.0489	96.30	0.3667
B	0.1192	100.00	1.3014	0.1192	100.00	1.3014
D	0.0361	100.00	0.3417	0.0361	100.00	0.3417

The old undercoverage is not reproduced in this unused-catalogue batch, but the revised uncertainty is conservative rather than sharply calibrated: nominal 68.27% intervals contain 94.44% of primary targets, and image B has mean 95% width 1.3014 mag. The no-driver secondary forecast retains mean signed residual +0.1761 mag. Moreover, the zero-delay control has mean 95% interval score 0.6334 mag, compared with 0.6779 mag for conventional, while conventional has the larger joint log density. Coverage, width, bias and the different proper scores must therefore be considered together; no uniformly superior model or precise delay recovery is established by this batch. The conventional intervals contain 160 of 162 primary targets (98.77%). The conditional R1-minus-conventional joint log-score difference is -3.460×10^{-5}; this is a finite-model predictive comparison, not a Bayes factor from a completed physical-state posterior. The fixed-body R1 activation has a primary log-score difference of 1.507×10^{-4} relative to conventional, with the same optical nuisance parameters. It isolates the original scalar response; the separately refitted R1 comparison also includes development nuisance and optimizer movement and cannot be attributed entirely to that activation. The matched conventional effective-mass representation has the same retained optical transfer as conditional R1 and therefore the same score by construction; it is not an independent confirmation.

Without the C driver, the conventional all-image forecast has joint log density 390.118, RMSE 0.2170 mag, 95% coverage 100.00%, mean interval width 1.2534 mag and mean interval score 1.2534 mag. The secondary R1-minus-conventional log-score difference is 2.497×10^{-6}. Primary and secondary targets overlap and cannot be counted as separate confirmations.

The experiment tests a fixed, calibrated finite nuisance ensemble. Its short chains have maximum basic $\widehat{R} = 5.19$, so the mixture is not a demonstrated converged posterior and supplies no complete physical-state marginalization. These LCO catalogue values were unused in model selection and arise from a separate monitoring campaign. Their retrospective difference-imaging reference includes 2022 data, however, so photon-disjoint calibration and validation are not established. Common reduction errors and prior qualitative publication awareness limit blinding. The central-image HST result is conditional on the empirical PSFs, adopted marginal noise and macro-to-observed flux relation. These qualifications, and any remaining undercoverage in the actual test, remain part of the result.

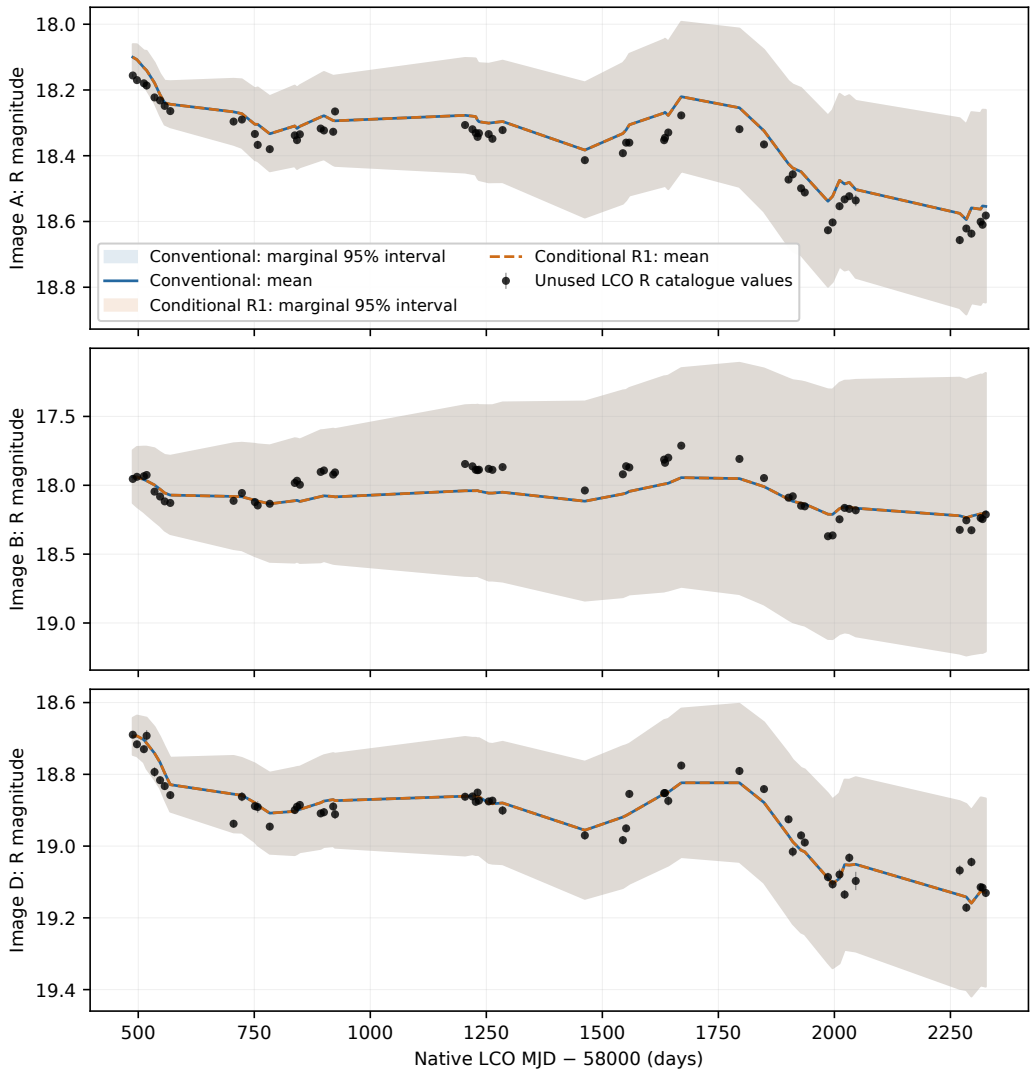


Figure 4. Unused LCO R observations and the frozen C-driven conventional and conditional R1 predictions. Shading shows marginal 95% mixture intervals, including the development-only image calibration. The curves and bands can nearly coincide. Straight segments join predictions only at observed epochs and are not an additional fitted variability curve.

Neither the statistical ensemble nor a close conventional/R1 score establishes a unique envelope state or a BHSM detection; the physical guard remains 2×10^{-10} and no physical admission is asserted.

9.5. A separate unused Gaia astrometric test

The LCO common-reference limitation motivates a second type of observation: the four image positions in Gaia DR3 [20]. We declare this additional test after the LCO result is known, while retaining exactly the already-frozen conventional and R1 physical predictions. A count-only catalogue query establishes four entries in the lens field. Numerical positions, errors and quality rows remain unopened until a separate prediction/protocol commit. No mass, screen, source, angle, plate scale or rotation is fitted against Gaia.

In a common small-angle ICRS tangent frame the three relative observables are $z_i = ((\alpha_i - \alpha_A)\cos\delta_A, \delta_i - \delta_A)$, $i = B, C, D$, with degree-to-arcsecond conversion. All 24 source assignments are compared with the fixed conventional relative pattern; the same assignment is used for R1 and its runner-up gap is retained. For each Gaia source, the published RA tangent-plane error $\sigma_{\alpha*} = \sigma_\alpha \cos\delta$, declination error and their correlation

define C_i. The RA error receives no second cosine. The full six-coordinate formal covariance is

$$[C_{\rm rel}]_{ij} = \delta_{ij} C_i + C_A, \qquad C_s = C_{\rm rel} + s^2 I_6, \quad s = 0, 2, 5 \text{ mas}. \tag{47}$$

The added floors apply per relative coordinate after differencing; they are prespecified sensitivity controls, not fitted noise or a derived macro posterior. The Gaussian density and quadratic residual use C_s, including its determinant and common-A off-diagonal blocks. Between-source catalogue covariances are unpublished, so source-error independence remains a conditional approximation. All four sources are retained irrespective of their quality diagnostics; missing mandatory errors cannot be replaced silently with zero correlation.

Gaia collected different photons in 2014–2017, whereas the fitted NIC2 point astrometry was published in 2011. However, the reprocessed HST central mosaics use a Gaia-EDR3 field frame. Their pixel test uses relative registration, but independence of every field calibration is not certified. Close-image blending, chromatic photocentres and fixed macro parameters further limit a catalogue-only Gaussian interpretation. This test assesses the previously unused numerical quasar geometry; it is not a complete Einstein-ring reconstruction or a BHSM state measurement.

The Gaia prediction freeze was committed before the numerical query. The unique closest A/B/C/D assignment has squared residual 6.3594×10^{-5} arcsec^2; the runner-up is 4.6197 arcsec^2. No continuous frame parameter is fitted. The conventional observed-minus-predicted east/north residuals are $(0.623, 1.701)$ mas for B, $(1.915, -5.026)$ mas for C and $(-5.572, 0.580)$ mas for D. Their coordinate RMS is 3.25560 mas, versus 3.25557 mas for R1. Table 10 reports every prespecified covariance control.

Table 10. Previously unused Gaia DR3 relative astrometry, with all physical predictions fixed before the query. $Q = r^T C_s^{-1} r$ uses six coordinates; the log density is normalized in mas^{-6}. Added floors are sensitivity controls, not fitted errors or a completed macro posterior.

Floor (mas)	Conventional Q	R1 Q	Conventional log p	R1 log p
0	3465.905	3465.667	−1727.177	−1727.058
2	15.7966	15.7963	−17.5910	−17.5908
5	2.54112	2.54107	−16.4441	−16.4440

Both frozen point predictions fail the catalogue-error-only precision check. The source RUWE values are 1.00–1.11, none is marked duplicated, and multi-peak fractions are 0–1%. B and D nevertheless have odd-window fractions of 64% and 69%, and C has excess-noise significance 2.57; these diagnostics remain visible without post-target image exclusion. Adding the declared floors reduces the formal mismatch, but cannot turn a sensitivity calculation into successful validation or establish the required physical uncertainty. The R1-minus- conventional log densities are 0.1191, 1.53×10^{-4} and 2.43×10^{-5} for the three controls. A small difference between inadequate point predictions is not evidence for a BHSM state. The next physical inference requires justified training/frame/photocentre and macro-model uncertainty together with richer mass/environment constraints. These Gaia targets are now consumed; fitting them later cannot count as another unused-data success.

9.6. Diagnosing and revising the restricted point geometry

The failed Gaia experiment in Section 9.5 now supplies development data, rather than another validation opportunity. Its discrepancy is largely present before the Gaia comparison: the frozen conventional model misses the NIC2 coordinates used for its fit by 3.0913 mas coordinate RMS, whereas Gaia and those NIC2 measurements differ

by 1.2469 mas RMS. Exact gnomonic projection changes the old comparison by at most 0.00322 mas. Fitting a global rotation and scale reduces the Gaia residual only from 3.2556 to 3.2393 mas. The image labels, east/north convention, common-A covariance and original stationary-ray residuals therefore do not explain the principal failure.

Nor does more optimization within the same declared family resolve it. Using all nine original parameters, the unchanged Gaia covariance and no delay or centroid penalties gives a local best quadratic residual of 1559.08, with 2.9861 mas coordinate RMS. The radial scale approaches its lower bound. In the eigenframe of the constant symmetric effective map, a circular deflector obeys $\beta_j = u_j[1 - g_j k(r)]$, where $u = \theta - c$ and β is the center-shifted effective source in that eigenframe. Eliminating the radial deflection factor gives

$$(g_1 - g_2)u_1 u_2 + g_2 \beta_1 u_2 - g_1 \beta_2 u_1 = 0. \tag{48}$$

The radial profile imposes further restrictions. Near the fitted old family the six astrometric rows have five strong derivative directions and one very small direction. These numerical local fits and the structural restriction motivate a less restricted geometry; they do not prove that every circular profile is excluded or identify a unique physical cause.

An independently documented candidate is the nearby galaxy G1 of Sluse et al. [21], called G22 in the earlier astrometry [22]. It lies at observed east/north offset $(-2.585, -3.637)$ arcsec relative to the main galaxy's light center and redshift $z_p = 0.7821$. The earlier paper calls the *main* lens G1; these aliases refer to distinct objects. We replace the wholly quadratic rear potential by a positive regular Plummer perturber plus a separately specified quadratic remainder,

$$\psi_p(x) = \tfrac{1}{2} x^T U_{\rm rem} x + 2 m_G \log \frac{|x - c_G|^2 + a_G^2}{2 a_G^2}, \qquad m_G = G M_G / c^2, \tag{49}$$

$$\Sigma_G(R) = \frac{M_G a_G^2}{\pi (R^2 + a_G^2)^2} \geq 0, \qquad a_G = 0.2 \text{ kpc}. \tag{50}$$

The fixed scale regularizes a distant point-mass surrogate; it is not a measurement of G1's internal profile. Its observed direction must itself be traced through the foreground lens. With the same distance convention as the original calculation, the actual equations evaluated are

$$x_p(\theta) = D_{op}\theta - D_{lp}\alpha_l(D_{ol}\theta), \qquad c_G = x_p(\theta_G), \tag{51}$$

$$R(\theta) = \theta - \frac{D_{ls}}{D_{os}}\alpha_l(D_{ol}\theta) - \frac{D_{ps}}{D_{os}}\nabla\psi_p(x_p) - \beta_0 = 0, \tag{52}$$

$$J = I - \frac{D_{ls}D_{ol}}{D_{os}}H_l - \frac{D_{ps}}{D_{os}}(U_{\rm rem} + H_G)(D_{op}I - D_{lp}D_{ol}H_l). \tag{53}$$

The Hessians in the last product need not commute. Both independent stationarity equations are also checked in the full arrival action

$$ct = \frac{k_l}{2}|\theta - \theta_p|^2 + \frac{k_p}{2}|\theta_p - \beta_0|^2 - (1 + z_l)\psi_l(D_{ol}\theta) - (1 + z_p)\psi_p(D_{op}\theta_p),$$
$$k_l = (1 + z_l)\frac{D_{ol}D_{op}}{D_{lp}}, \qquad k_p = (1 + z_p)\frac{D_{op}D_{os}}{D_{ps}}. \tag{54}$$

Setting $M_G = 0$ at the old rear redshift reproduces the previous conventional and original R1 image and delay applications. The new potential introduces spatial variation of the tide, not a fitted image-specific displacement or an uncertainty floor.

The conventional and conditional R1 fits use the consumed Gaia positions, the old lens-light position, the prescribed 3 mas mass-center prior and the original development-light

likelihood. Exact gnomonic coordinates and their six-by-eight origin-dependent covariance propagation replace the small-field coordinate approximation in this new fit. The quadratic *remainder*'s convergence is anchored to the inherited value to fix a mass-sheet degree; it is not an independent measurement of the new total convergence. Nine parameters remain free. The development Gaia quadratic residuals become 0.02651 and 0.02628, with coordinate RMS 0.01055 and 0.01051 mas, respectively, using unchanged catalogue errors. Six relative coordinate values cannot uniquely identify these parameters. Weak nuisance directions and different numerical optima are retained in the receipt; the improvement is not a validation or proof that G1 is the unique omitted response. The earlier LCO scores and coverage belong to their earlier frozen models. They are not assigned to this revised G1 geometry or counted as a fresh test of it.

A controlled counterfactual isolates the spatially varying rear tide on each final frozen body. At the declared rear origin $x_0 = 0$, replace only the G1 potential by

$$\psi_G^{(2)}(x) = \psi_G(0) + \alpha_G(0) \cdot x + \tfrac{1}{2} x^T H_G(0) x. \tag{55}$$

The constant potential, constant deflection and full Hessian agree at that origin; U_{rem}, the local body, source, G1 center, distances and clock remain fixed. Solving the four image branches in the same original two-plane equations changes relative coordinates by up to 22.56 mas and relative delays by 1.23×10^4 s in either model. The Taylor branches satisfy the ray residual below 3×10^{-21} rad and the independent arrival gradients below 1.4×10^{-17} Mpc. On the already consumed Gaia coordinates, the fixed-body Taylor comparison has about 14.08 mas coordinate RMS, whereas the full G1 fits have the small development residuals above. No parameter or uncertainty is refitted in this comparison. The original G1 Hessian has positive trace 3.36×10^{-9} Mpc^{-1} at the expansion origin, consistent with $8\pi G\Sigma_G/c^2$ within 1.4×10^{-12} relative error. This identifies a substantial nonconstant optical contribution within the prescribed frozen model. The Taylor polynomial is a local counterfactual, not an independently completed finite-mass source; this comparison neither establishes G1 as the unique physical cause nor validates predictions on unused observations.

Both new finite image searches find seven verified stationary images. Their central/A macro ratios are 0.00165054 and 0.00165038; the existing conditional F814W application permits both, with upper scale multipliers about 7.46. Faint additional images near G1 remain in the census. The original local nonlinear scalar equation still consumes the *total* two-Plummer enclosed mass once, on its original regular branch. Its sections and the full two-plane arrival equations are checked on each matching application. The final sampled scalar and metric Gauss section residuals are below 7×10^{-16} and 4×10^{-16}, respectively. Independent deflection- and potential-derivative checks instead differ by 1.9×10^{-12} and 3.6×10^{-9}; these distinct diagnostics are not all machine-precision checks. Finite projection and outer matching limitations remain recorded in the application receipt. G1 is a prescribed positive *total optical* potential common to the comparison, not a newly solved BHSM environmental scalar state. Material support, global matching and the microscopic preparation dependencies remain as stated above; the 2×10^{-10} physical guard is unchanged and physical admission remains false.

9.7. A separately frozen native JWST geometry experiment

The new observations are the four 2022 October 12 JWST/NIRCam F115W NRCB3 native calibrated exposures in program 1198, totaling 1803.776 s. These are different telescope/detector observations from the consumed HST and Gaia data. Header-only inspection selected the detector before any science array was acquired. The physical predictions, the four-file choice, the field-calibration algorithm and the statistical criteria were

committed at `fc3a449` before native-file acquisition. Gaia copies, alternative reductions of consumed HST photons and the earlier LCO tests are not fresh targets.

The measurement uses native SCI, ERR and DQ arrays and the embedded official nonlinear instrument GWCS. It does not replace that map by a linear FITS WCS or fit an affine correction to the quasar configuration. A ten-arcsecond sky guard excludes the quasar region from field calibration. The initial empirical-PSF builder finds only two eligible candidates per exposure, below its declared three-training-star requirement; the known 270-pixel dither steps prohibit pooling a common PSF. A serialization failure and its calibration-only rerun are preserved separately.

A subsequent, separately committed calibration uses the independent STScI LMC F115W NRCB3 standard-PSF library, sampled at each native source position [23,24]. The official reader's sampling, origin, spatial interpolation and orientation are verified against the actual library. Gaia field source 3178020716638520576 supplies an independently identified star in all four footprints. Its saturated core is masked. The unsaturated wings supply a field-only translation of approximately $(-18.25, -28.27)$ native pixels for association initializers; this does not change the measured scale or shear. Every 17-by-17 field fit is interior with rank six and 268–273 usable pixels, but its known-ERR quadratic residual is 11381–14090 for 262–267 degrees of freedom. All four fail the fixed $p \geq 0.001$ adequacy gate. The native wings exceed the reference profile by about 10–23% near radius 4–5 pixels. Most discrepancy comes from unflagged pixels. Positive Gaussian jitter and halo convolutions improve the mean only modestly and remain inadequate. Their exploratory fitted corrections are not transferred to the quasar response.

The source-specific instrumental response has the general defining form

$$P_{ie,p} = \frac{1}{\mathcal{N}_{ie}} \int \mathrm{d}\lambda \, S_i(\lambda) T_e(\lambda) \int_{\Omega_{ep}} \mathrm{d}^2 u \left[\left| \mathcal{F}\{A_e \exp(2\pi i W_e/\lambda)\} \right|^2 * K_{J,e} * K_{D,e,\lambda} \right](u), \quad (56)$$

where S_i is the incident photon spectrum, T_e the throughput, A_e, W_e the pupil amplitude and wavefront, K_J, K_D the image-motion and detector responses, and Ω_{ep} the native pixel. The pupil is evaluated at the source's field position; u is relative to its trial centroid, with the native pixel domain shifted accordingly. The normalization absorbs the independently fitted flux. A fixed empirical library specifies one effective instance, not every source spectrum or visit response. The field residuals show that the tested instance is inadequate; they do not uniquely identify the wavefront, spectral or detector cause. The already integrated ePSF is not integrated a second time, and native CAL fitting follows the official consumer's direct SCI/ERR convention. A calibrated precision extraction requires an adequate instrumental mean and its uncertainty, not a larger error bar.

Before opening any quasar values, a further committed amendment permits an explicitly instrumental diagnostic with the original fixed library PSFs. The original precision-validation experiment remains blocked. Every validation flag in the diagnostic is forced false, including when raw geometry improves. A quadratic smooth background is specified using already consumed HST curvature diagnostics and independent synthetic injections, before JWST target fitting. Neither the PSF, physical lens parameters, point masks nor native errors are tuned to these targets.

The scale-preserving comparison removes only translation and rotation. Writing $u = (B - A)/|B - A|$ and v for its positively oriented normal, the five measured coordinates are

$$q = \big(|B - A|, (C - A) \cdot u, (C - A) \cdot v, (D - A) \cdot u, (D - A) \cdot v \big). \quad (57)$$

The assembled conditional eight-by-eight centroid covariance is pushed through the instrument map and the five-by-eight derivative, retaining the shared A and AB reference terms. Formal centroid information is not multiplied by the observed residual variance. Fixed

17-by-17 pixel PSF fits and the four-exposure consistency test must each pass their declared adequacy gates. Only then can the frozen five-coordinate point prediction be assessed at level 0.05. A claimed improvement additionally requires at least a 25% reduction in raw coordinate RMS relative to its matching old prediction on the same data. No scale fit, residual-tuned floor, broader-interval success, disagreement-based image exclusion or switch to a reserve band is allowed. These remain conditional tests of point geometry: finite field-PSF uncertainty and shared distortion covariance are not supplied by ERR and do not constitute a fully marginalized astrophysical posterior.

The one-time diagnostic opens all four declared exposures and completes all sixteen point fits. The extraction fits free centroids once; no prediction-constrained second extraction, target scale or physical-lens parameter fit is applied. The fixed prediction residuals are shown in Table 11. The conventional raw five-coordinate RMS falls from 3.31331 to 0.779400 mas, a 76.48% reduction; R1 gives 3.31316 to 0.779454 mas. This is a reduction in the residual itself, under identical errors and data. It is not an interval-width success.

Table 11. First application to previously unused native JWST F115W quasar pixels. The five-coordinate residual RMS is descriptive. The quadratic statistic uses the fixed conditional centroid covariance; failed PSF and exposure gates prevent interpreting it as a calibrated physical-model test. Every validation flag is false.

Frozen prediction	Raw q RMS (mas)	Formal Q_5	Validation
Old conventional	3.31331	18667.78	No
Old R1	3.31316	18667.23	No
G1 conventional	0.779400	958.434	No
G1 R1	0.779454	958.428	No

All sixteen target pixel fits fail the unchanged $p \geq 0.001$ gate, with quadratic residual per degree of freedom from 24.72 to 79.97; the optimizers are interior. The four native configurations also fail the model-independent common-geometry check, $Q_{\mathrm{scatter}} = 432.262$ for 15 coordinates ($p = 1.13 \times 10^{-82}$ under the conditional errors). For $W_e = C_e^{-1}$ and $\bar{q} = (\sum_e W_e)^{-1} \sum_e W_e q_e$,

$$\sum_e (q_e - q_*)^T W_e (q_e - q_*) = \underbrace{\sum_e (q_e - \bar{q})^T W_e (q_e - \bar{q})}_{Q_{\mathrm{scatter}}} + (\bar{q} - q_*)^T \Big(\sum_e W_e \Big) (\bar{q} - q_*). \quad (58)$$

A shared lens prediction cannot remove the first term. Thus the G1 correction reduces the raw diagnostic residual on newly accessed photons. Uncalibrated centroid bias prevents identifying that reduction with an improvement in the true physical geometry or precision validation. Neither another common lens displacement nor a claim based on broader errors resolves this instrumental consistency failure. The sub-milliarcsecond remainder cannot yet be assigned to a BHSM interface or envelope state. An adequate visit- and source-dependent PSF/detector response, diffuse host model and distortion/noise treatment are needed to separate those effects. The tested response corrections have not supplied that solution. No remaining band is opened to replace the failed primary check, and no target-refitted geometry is reported as fresh validation. The physical guard remains unchanged.

9.8. Rate eligibility and read-history-aware response development

The preceding first extraction contains a specific pixel-selection error. For the archived native JWST/NIRCam `CAL_VER=2.0.1` rate products, `SATURATED` can report saturation in a later group while earlier groups still supply a usable slope [25]. Rejecting every

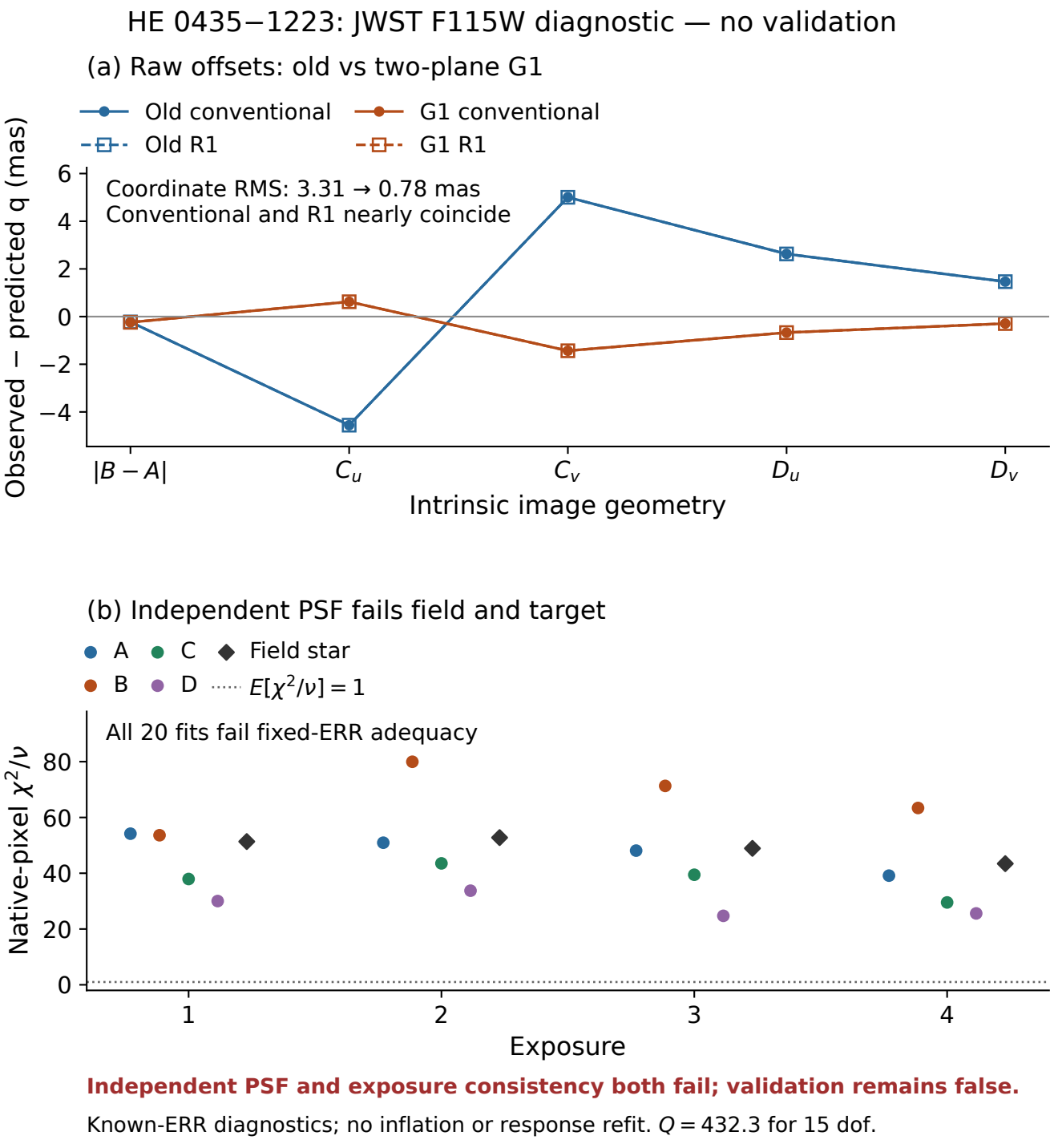


Figure 5. Unused-photon geometry diagnostic and native pixel adequacy. Fixed G1 predictions reduce the raw geometry residuals. The severe pixel residuals independently prevent validation; formal centroid errors are not inflated. Conventional and R1 curves nearly coincide.

such rate discards information. An additive, release-checked adapter retains a slope only when SCI and ERR are finite, ERR is positive and `DO_NOT_USE` is absent. It checks native two-dimensional ImageModel/rate provenance and completed ramp fitting; it does not apply this rule to raw GROUPDQ, mosaics or unverified releases. The original fitter, SCI/ERR/DQ arrays and frozen checkpoints are preserved.

The supplied repair package cited the correct immutable source revision but carried a legacy-fitter digest inconsistent with that revision. Its default guard rejected the replay before data application. A separate integration manifest binds the actual original Git object and the same ten JSON inputs; it changes neither the fitter nor any physical prediction. The delivered files and failed guard remain archived. Canonical native-coordinate replay then reproduces all sixteen original centroids exactly in the current runtime. This is an explicit provenance correction, not a silent relaxation of an input check.

The verified adapter restores 303 valid slope pixels. With the original fixed library response, the G1-reference raw q RMS becomes 0.448457055 mas and the R1-inclusive value 0.448562655 mas. The exposure statistic becomes 253.027695 for 15 coordinates. All sixteen absolute profile checks still fail, with quadratic residual per degree of freedom 27.57–83.55. Restoring information tightens conditional centroid errors: the G1-reference formal geometry statistic is 1033.114 for five coordinates despite its smaller raw discrepancy. The original 0.779400 mas result remains the historical first extraction; the correction is a development reanalysis of the same photons.

The package also contains completed response trials, preserved as delivered rather than rerun here. A positive field-trained response transfers partly, giving about 0.362 mas raw RMS, but held field and target profiles fail. Finer response bases, limited spatial gradients, local-quasar calibration, dated optical color/diffusion and eight bounded pupil-phase modes also remain inadequate. Those prior trials do not authorize a new arbitrary PSF family or a coefficient fitted to the gravitational residual.

The delivered native detector-history calculation supplies a more specific lead. A numerically compatible `stcal 1.18.0` ramp fit reproduces the archived field rate and ERR within 8.97×10^{-8} and 1.65×10^{-7} relative, respectively. The archived installation version is not recorded; numerical agreement is not proof of that installation history. The earlier incompatible replay is retained. No completed native detector processing is repeated for this integration. On exactly 289 matched field-star pixels, a separately calibrated first-two-read difference reduces normalized raw profile residual from 7.6653% to 5.1787%. Its $p = 0.00984653$ still fails the declared 0.01 gate, formal centroid errors grow about threefold, and several wing annuli worsen. Neither the lower quadratic statistic nor a shorter common-time extraction is promoted to validation.

The image operator must therefore expose the consumed histories and extended light. For a conditional BHSM sky and instrumental parameters η, the native pixel mean has the evaluated structure

$$I(\theta,\lambda) = \sum_i F_i(\lambda)\delta^2(\theta - q_i) + \sum_k a_k(\lambda)B_k(\theta),$$
$$f_{ep} = \int \mathrm{d}\lambda\, T_e(\lambda) \int \mathrm{d}^2\theta\, P_{ep}(\theta,\lambda;\eta) I(\theta,\lambda), \tag{59}$$

followed by the read-accumulation and slope-estimation operator. Here P_{ep} already includes the native pixel integral. The sky quadrature in this equation integrates over source position and does not integrate that pixel response a second time. The frozen optical map supplies the physical predicted point positions; during response development the measured q_i are independent nuisance coordinates. Agreement with a lens prediction is never the calibration objective. The shared observed-sky basis describes only the sampled

neighborhoods; these disjoint stamps do not uniquely reconstruct galaxy and lensed-host light over the full field.

With the actual individual read times t_{jr}, we evaluate the following leading time/illumination candidate:

$$\mathbb{E}[G_{pj}] = a_{p,s(j)} + \langle t \rangle_j f_p + \frac{\kappa_e}{2g_p} \langle t^2 \rangle_j [L((gf)^2)]_p, \tag{60}$$
$$\mathbb{E}[\widehat{f}_p] = f_p + \frac{\kappa_e}{2g_p} \sum_j w_{pj} \langle t^2 \rangle_j [L((gf)^2)]_p.$$

The symmetric neighbor-exchange graph L has zero total flux, so this candidate conserves charge at each read on its specified domain. Pixel-specific slope weights can nevertheless yield a nonconserved sum of the final slopes; renormalizing those slopes would hide the history dependence. DNU and SAT exclude a group, while a jump starts a new intercept segment. The actual group average of t^2, rather than the square of the group mean time, is required. For static linear response, the group-estimator identities $w_p^T H_p = 0$ and $w_p^T m = 1$ eliminate segment intercepts $H_p a_p$ and return f_p; different histories alone cannot alter a static linear profile. The second line of Equation (60) is this linear group-estimator identity. Exact archived OLS_C weights and individual ZEROFRAME branches are not reconstructed or applied; the actual new likelihood fits the corrected groups directly. This candidate is a specified leading response hypothesis, not a uniquely derived charge-migration law or a measured detector constant. In this direct-group application f_p is in corrected DN s^{-1}, g_p is the recorded gain in electrons per DN and κ_e is in electron^{-1}. The graph acts on accumulated electron charge; conserving DN alone would not conserve charge when the gain varies. Fixed flat and photometric maps convert the optical surface-brightness prediction to the count units. An initial DN-conserving surrogate is retained as a superseded development implementation, with its unit correction explicit. The delivered gain is spatially constant at 2.05 electrons/DN, so its predicted means coincide with the corrected electron formulation after rescaling the coefficient. A nonuniform-gain regression checks the general conservation law; no electron leakage is inferred for this particular constant-gain data set.

Shared reads also require temporal covariance. With the reference CDS read-noise convention [26], the conditional photon term for two two-read groups is

$$C^{\mathrm{ph}}_{p,jk} = \frac{f_p}{4g_p} \sum_{r,s} \min(t_{jr}, t_{ks}), \qquad C^{\mathrm{read}}_{p,jk} = \delta_{jk} \frac{R^2_{\mathrm{CDS},p}}{4}. \tag{61}$$

The common-read covariance must enter a held-group prediction, instead of summing early and full-rate likelihoods as independent data. Linearity derivatives, SIRS-induced spatial covariance, dark/reference covariance and observed-selection effects remain distinct limitations. Moreover, the stipulated mean exchange does not specify a stochastic electron-transfer law, so it does not determine redistribution-induced interpixel covariance. These are conditional mean applications with a frozen working covariance, not a complete detector likelihood.

Learned optical-response uncertainty also induces correlations between image centroids. With one declared SVD convention on a whitened, dimensionless Jacobian, the local joint information must profile all shared response and sky nuisance directions together. A fixed-PSF two-by-two inverse information per image omits those correlations. All resulting uncertainty statements remain conditional when the mean profile is inadequate. The following new applications assess this functional structure on consumed development

observations; none opens a reserved band, repeats the original blinded experiment, or changes the 2×10^{-10} physical guard and false admission state.

The first executed temporal application jointly fits the first eight groups with independent pixel rates and segment intercepts, using the ninth group only for a conditional prediction. Its working covariance is frozen from training-only rates and includes the shared-read term. The ninth group was already present in the delivered observations; this is an internal temporal holdout, not a new unused-photon test. The fit gives $\kappa_e = 2.3627 \times 10^{-7}$ electron^{-1} with local conditional curvature error 3.3518×10^{-8} electron^{-1}. Training Q falls from 1621.402 to 1566.524. On 257 retained held pixels, conditional-prediction Q falls from 272.942 to 260.747 and raw residual RMS from 10.20884 to 9.67245 DN, a 5.25% reduction. Free pixel rates make this a temporal-curvature diagnostic rather than optical-profile calibration. Missing calibration/spatial covariance prevents interpreting the curvature error as a complete detector-coefficient measurement.

An actual coupled optical test then restricts the rate map to the fixed dated 4000 K, equal-sampling STPSF [27], free centroid and flux, and one background plane. These six optical nuisance coordinates are jointly fitted with κ_e through the original group operator. The full seven-parameter local covariance retains their correlations. Although training Q improves from 11909.177 to 11569.456, the held group worsens: Q rises from 1542.413 to 1573.909 for 257 pixels and raw residual RMS from 25.23791 to 25.58811 DN. The fitted conditional $\kappa_e \simeq 2.3665 \times 10^{-7}$ electron^{-1} therefore does not repair the optical mean. This controlled failure distinguishes a measurable temporal lead from a sufficient PSF response; no such coefficient is transferred to quasar extraction or assigned to a gravitational term.

The separate rate-image application evaluates Equation (59) with a coherent sky. Each of the four disjoint image neighborhoods has a 3×3 basis of nonnegative observed-sky Gaussians of width 0.10 arcsec and spacing 0.18 arcsec. Its 36 coefficients are common to all exposures, with a declared zero-centered Gaussian penalty of scale 10 in SCI units restricted to nonnegative coefficients. Exact archived GWCS maps sample that sky at one quarter of a native pixel. The already pixel-integrated response is convolved once, with zero response outside its declared 25-pixel support and padded sky quadrature; the likelihood uses the canonical 17×17 native stamps. One detector background plane per exposure replaces independent quadratic backgrounds around each image. Point coordinates and fluxes remain free measurement nuisances. No physical lens prediction is loaded by this calculation.

The response retains the delivered positive field family with 318 coordinates:

$$P(u;\eta) = \frac{\mathrm{spline}(P_0)(u)\exp[B(u)\eta]}{Z(\eta)}. \tag{62}$$

Its original gauge removal and finite-grid normalization are retained. Four field-star stamps supply the calibration likelihood; the declared coefficient prior has scale 0.2. The objective evaluated retains the original native residuals

$$r_{ep} = \frac{d_{ep} - f_{ep}(q,F,a,\eta) - b_{e0} - b_{ex}x_p - b_{ey}y_p}{\mathrm{ERR}_{ep}},$$
$$Q = \sum_{ep} r_{ep}^2 + Q_{\mathrm{field}} + \|a/10\|^2 + \|\eta/0.2\|^2. \tag{63}$$

Field flux, position and affine background are profiled at each evaluated response. Native ERR is unchanged. This sky is a finite observed-light nuisance description; its positive patches do not identify the separate galaxy and source-plane host components.

First, field-only calibration uncertainty is propagated at the delivered response. If A and B are the whitened point/sky nuisance and response Jacobians and F_p includes

the stated sky penalty, the cut covariance is $C_p = F_p^{-1} + SC_{\eta,\mathrm{field}}S^T$ with $S = -F_p^{-1}A^TB$. This retains common calibration uncertainty without allowing target pixels to update its mean. A separate coupled Gauss–Newton application then updates point, sky and response coordinates together, re-profiles the field nuisances, and accepts a step only after re-evaluating the original nonlinear Equation (63). All ranks use singular values of dimensionless columns above 10^{-10} times the largest value. Null projections and the distinction between data rank and prior-supported rank are recorded.

One actual coupled step lowers target raw Q from 50121.532 to 34712.081 on 4620 pixels; the complete field-plus-target-plus-prior objective falls from 57432.279 to 45065.902. It is an executed continuation iterate, not a converged fit. The largest unconstrained scaled residual gradient is 14477 and includes active-bound directions; it is not a projected convergence statistic. A separate saved-mean evaluation gives a free detector-plane scaled half-objective gradient of 12254 and projected nonnegative-sky violation 7605, establishing a remaining stationarity failure. Thirteen sky coefficients are on their zero bound. The 414-column local joint problem has rank 414, but its covariance is unconstrained local curvature at this nonstationary iterate, not a truncated or calibrated posterior. The full 32×32 centroid covariance and 20×20 intrinsic-geometry covariance include shared-response, sky and cross-exposure terms. The largest absolute off-diagonal centroid correlation is 0.333. Raw centers, full finite-grid response moments and baseline-relative moments are retained separately; no finite-grid convention establishes physical centroid accuracy.

Figure 6 retains the actual native residual pattern at this saved iterate, rather than displaying only a reduced objective.

A second application trains the shared sky and response on three target exposures, retaining all four field-star calibrations. The fourth target exposure is then evaluated with only its independent point/flux and detector-plane nuisances free. Its native residual is $Q = 10906.483$ for 1141 nominal conditional degrees of freedom, and fails absolute adequacy. The full trained sky/response covariance, including their cross block, is propagated to its centroids; it is not used to inflate ERR or turn the failed profile into a pass. This whole-exposure check uses already consumed photons and a provisional training iterate. It is neither a fresh physical validation nor a definitive rejection of every converged coherent-sky model.

The concrete remaining measurement dependency is thus an adequate calibrated mean together with its stochastic and reference covariance. The compact supplied read-history array covers one field star in one exposure, rather than all quasar groups and selections. Those actual histories are required for a joint target history/sky application; their absence from that compact delivery is not evidence that they do not exist. The failed optical transfer already prevents exporting the fitted κ_e to the target images. Neither substituting an unmeasured target history nor assigning the field residual to an extra envelope amplitude supplies this dependency. The microscopic state/source and domain conditions stated in the Discussion remain separate from these instrumental limitations.

10. Computational Methods and Reproducibility

The calculation uses the fixed manuscript source revision `49a1dab` and records full input hashes. The main reproduction command from the repository root is

```
python scripts/run_strong_lensing_r1_screened.py
```

The coefficient, independent projection, and multiple-source checks are under `scripts/r1_screened_review/`. Numerical results and their units are under `artifacts/strong_lensing_r1_screened_20261020/`. The research package additions and exact commands are integrated with the pinned repository background and optical dependencies. The recorded package-era replay of three portable review scripts reproduced all 305 saved

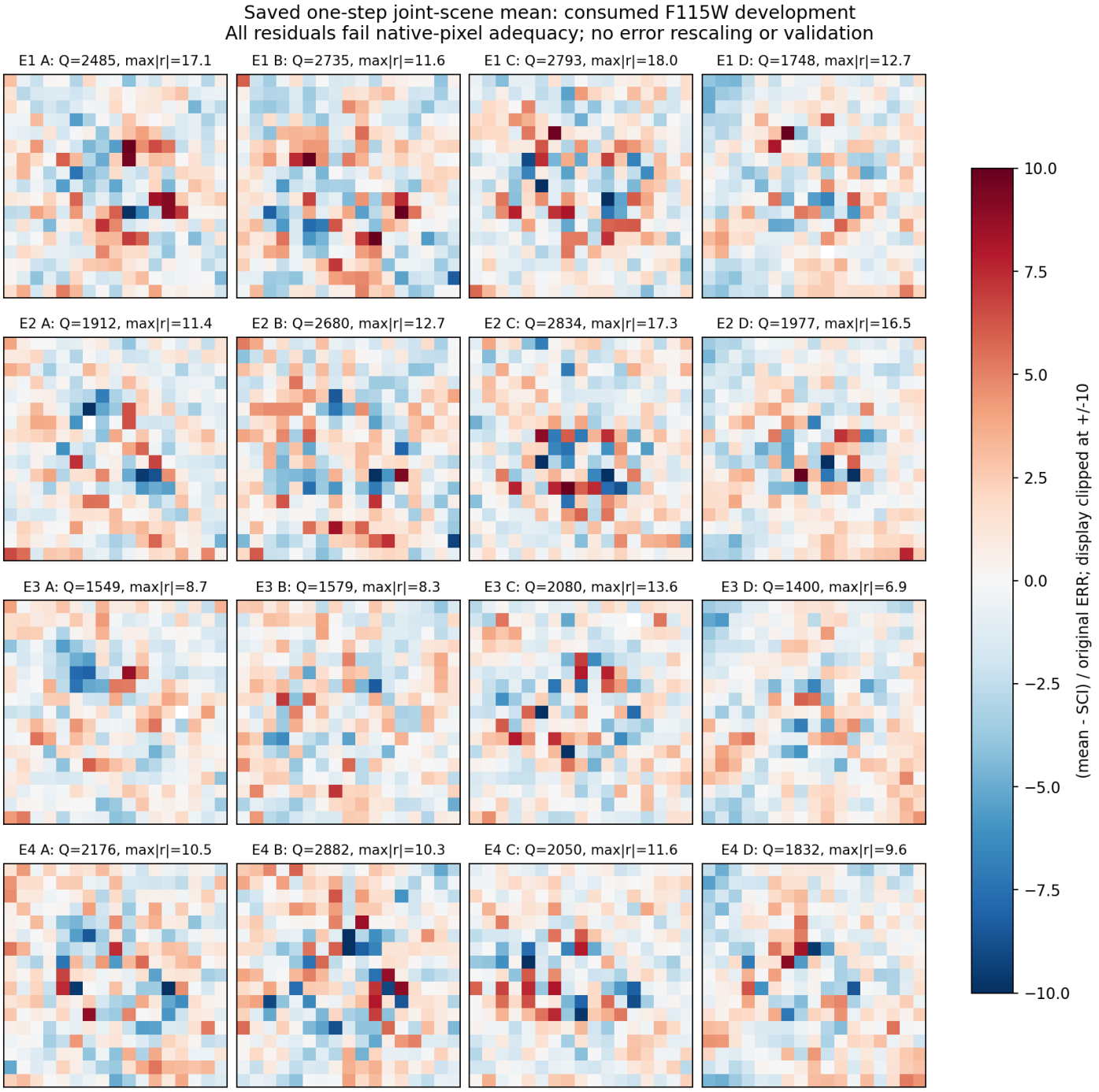


Figure 6. Native 17×17 standardized residuals after the one-step coherent-sky/response application. Rows are the four consumed exposures; columns are images A–D. Color shows $(\text{mean} - \text{SCI})/\text{ERR}$, the opposite sign to Equation (63). Colors alone are clipped at ± 10; raw arrays, quoted Q and maximum absolute residuals are not clipped. Excluded pixels are masked. All profiles remain inadequate; this is development evidence, not validation.

numerical values in that environment. The current integration replayed nine numerical commands, compared 1161 saved numerical values, and passed 16 focused checks without changing the original numerical artifacts. These are separate execution records, not independent physical measurements.

The additional inverse and profile applications are reproduced with `scripts/run_strong_lensing_refit.py` and `scripts/run_strong_lensing_robustness.py`; the distributed optical control uses `scripts/run_strong_lensing_encounter_accuracy.py`. Their outputs are under `artifacts/strong_lensing_acceptance_20261020/`. Four refit, seven profile/coefficient and seven encounter tests check the original applications and independent projections. The distributed-control receipt records its boundary identities, original variational residuals and convergence checks. The complete commands and software versions are in the acceptance checkpoint. The original 305-value portable package check and the current 1161-value replay remain separate records.

The reproducibility manifest identifies versions and hashes rather than equating successful execution with physical validation.

The independent quadrature uses the same physical radial equation but a separately implemented integration variable and root calculation. Its agreement is a reproducibility check, not independent empirical evidence or external human peer review.

The observed application archives original COSMOGRAIL/CDS files and row-level provenance, the pre-fit protocol, all four training starts, the committed pre-scoring predictions and the single reserved-target application. Eight light-curve and seven macro tests check exact likelihood normalization, dense conditioning, target-label invariance, delay sign/recovery, compatible maps, two-plane action reduction and consistent radial projection. The executed application and forward-audit receipts verify original scalar-source and metric-Gauss residuals. The full source and conditional-statistics definitions are in Supplementary Notes S9–S10. The scoring-start record persists even after an aborted exposure, so these targets cannot silently be reclassified as unseen after model revision. A separately labeled known-outcome replay reproduces the saved predictions and scores without refitting or reopening the experiment; it is not a second independent test.

The separate revision archives exact HST cutouts, all five compact-source candidates, their original scalar and optical applications, the six development windows and their calibration, and the committed LCO prediction freeze. The commands `scripts/fit_he0435_revision.py`, `scripts/fit_he0435_macro_revision.py` and `scripts/test_he0435_central_image.py` perform the development applications; `scripts/run_he0435_revision_validation.py` separates prediction preparation from the single guarded unused-target evaluation. The reporting script reads completed score receipts and performs no fitting. Independent dense-covariance and Python-filter comparisons check the optional C likelihood accelerator. Focused tests cover the normalized calibrated mixture, growing forecast variance, total nonlinear matter source, nonzero HST registration rotation, overlapping nuclear-image templates, and the persistent exposure lock. The numerical record distinguishes short-chain nuisance-ensemble uncertainty from a converged physical posterior. The additional Gaia protocol pins the already-fitted relative image predictions before its numerical catalogue query. Eight pure-mathematics and synthetic-runner checks verify coordinate signs, the RA-error convention, shared-reference covariance, labels, normalized density, missing-error handling and the exclusive query lock. No real target row enters those tests. The full focused application suite contains 81 tests; the 12 exact continuation identities retain their separate symbolic audit.

The subsequent geometry application is separately archived at calculation revision `7ad3cf175666`. It preserves the pre-acquisition physical freeze `fc3a449`, independent-library calibration freeze `29dcff9`, pre-opening diagnostic amendment `287b284`, the once-

only target marker, failed field diagnostics and original per-pixel residuals. Sixteen exact native SCI/ERR/DQ windows and four serialized official GWCS maps support portable inspection; public native product identities and full-file hashes recover the larger archive inputs. The fixed predictions are byte-identical to their pre-acquisition freeze. The additional geometry, pixel, calibration and guard checks bring the focused suite to 123 passing tests; internal post-opening checks reproduce the saved extraction and scores without refitting. Reproduction checks are not observational validation.

The additive response repair is documented in `docs/HE0435_RESPONSE_REPAIR_EXECUTION_20261021.md`, with its delivered byte manifest, failed initial source guard, explicit Git-bound provenance correction and canonical native replay. New read-group and joint-sky applications retain actual residuals, covariances, rank/bound diagnostics and superseded implementations under `artifacts/strong_lensing_response_repair_20261021/`. Independent reviews recompute held-group statistics and covariance transforms from saved arrays. The current focused suite has 146 passing tests, including the new mask, history and joint-scene checks; the 12 algebra identities retain their separate successful audit. These implementation tests do not repair failed empirical adequacy. The repair source, numerical applications and reviews are frozen at revision `34694083c22d`.

11. Discussion: State Inference and Physical Scope

The calculation answers one forward and inverse modeling question. A specified R1 state and finite source determine a regular encounter and its light observables. The coefficient derivatives quantify that state dependence. A positive effective mass represents the same retained metric, and a finite conventional nuisance fit reproduces the injected three-image system. Accordingly, the reference-relative seconds are a calculated response at fixed preparation; assigning them to a unique envelope state requires restrictions on the ordinary lens model and the state-to-source relation. A shared multiple-source profile can retain information that a one-source fit absorbs, but the complete positive effective-mass representation remains available at this order. This is a constructive equivalence within the BHSM transfer program.

The smooth-profile comparison gives the practical scale of this issue. The imposed profile changes alter ordinary delays much more than the state-conditioned offsets. They do not define an empirical prior; they show why a model-conditioned transfer is insufficient on its own for state identification. The finite refit is likewise a saturated local example rather than a forecast. The HE 0435 application supplies an actual but restricted test using point imaging and original brightness likelihoods. Its additional central-image prediction and conditional optical screen prevent interpreting that point fit as a complete measured galaxy or independently reconstructed environment.

The executed chronological test illustrates how existing light can establish a per-lens baseline, and how reserved observations can test it. Its conventional/R1 predictive tie is consistent with the effective-mass equivalence. Its brightness undercoverage, particularly in image A, shows why recovering a transfer does not establish a calibrated microlensing/source-variability model or unique physical state. The first experiment's two-stage point-fit intervals exclude parameter uncertainty. The separate revision uses a finite nuisance ensemble and development-only long-horizon calibration, then opens unused LCO catalogue targets once. The sample, driver and uncertainty rule all change, so the two coverage fractions are not a controlled single-variable ablation. Interval widths and proper scores quantify the cost of broader uncertainty. Coverage remains descriptive for correlated observations. No later residual is assigned to envelope evolution or a new interface amplitude.

A progressively constrained optical map needs the admitted mass, source, line-of-sight and calibration uncertainties and a specified interface law for an interface-state inference. A published cosmological posterior conditioned on a lens model cannot replace the light-data likelihood. For HE 0435 the HST-conditioned compact source and unused-catalogue test implement that next inference stage. HST constrains excess central light conditional on the macro-to-observed flux relation, while the LCO test assesses an empirical brightness prediction. Neither selects a unique composition for the compact source or supplies the global environmental scalar Hessian. The positive conventional representation remains available. This is the same transfer-to-identifiability question tested first in the controlled finite-body calculation.

The compact-source model's unused Gaia geometry gives a separate adequacy failure: both frozen point predictions have about 3.26 mas coordinate RMS and fail the catalogue-error-only precision check. The predeclared 2 and 5 mas floor controls expose uncertainty sensitivity; they are not an empirically justified replacement for training, frame, photocentre and macro-parameter uncertainty. Source-quality flags do not justify discarding the discrepant images. Development diagnosis finds that the old model already misses its fitted NIC2 geometry and that the small coordinate-map correction cannot explain the discrepancy. An explicitly positive G1 perturber supplies spatially varying tidal deflection absent from the quadratic screen. It reduces consumed-Gaia RMS to about 0.011 mas under unchanged errors. Nine free parameters and six relative coordinates make that fit underidentified. The frozen-body Taylor comparison isolates the nonlinear contribution within this model, without establishing a unique physical cause. Only the subsequent unused-observation application can test the revised prediction; the old LCO scores cannot validate it.

That new F115W application reduces raw five-coordinate RMS from 3.31 to 0.779 mas without changing errors or refitting lens parameters. The instrumental field-star response, all sixteen native pixel fits and exposure consistency nevertheless fail. The scatter identity in Equation (58) shows why a common lens correction cannot repair the exposure-dependent part. This result shows a smaller raw diagnostic residual for G1. Uncalibrated centroid bias prevents inferring that the true physical geometry is correspondingly improved; neither precision validation nor a BHSM preference follows. An adequate response in Equation (56) must precede a physical interpretation of the remaining sub-mas residual. The unused bands remain reserved, rather than replacing the primary outcome.

The subsequent rate-mask correction in Section 9.8 reduces the same-data raw RMS to 0.448 mas by restoring 303 usable native slopes. Its reduced raw residual does not override the failed profiles or the larger formal geometry statistic after conditional errors tighten. The delivered common-time history lead is then tested through actual electron-conserving group means: free pixel rates improve an internal temporal prediction, but coupling the same term to the dated optical profile worsens the held residual. A coherent shared sky and learned response covariance remove the assumption of independent fixed-PSF centroid errors; the executed nonconverged continuation nevertheless fails a held target exposure. Thus the calibrated instrumental likelihood remains an empirical dependency of envelope-state inference, distinct from the microscopic state-to-source map. No field-star residual is assigned to an extra gravitational response.

The approximation controls have separate meanings. The common GR2 term moves absolute image coordinates by up to 1.72 microarcsec, while its change of the reference-relative image offsets is below 7×10^{-6} microarcsec and its delay-offset change is below 0.1 ms. The distributed-encounter control tests the optical plane within the retained leading action. Its finite-window and path effects are distinct from scalar-dependent field-equation higher orders. The local parameter $H_l r_V / c = 5.64 \times 10^{-5}$ and the 1.1–1.7 $\times 10^{-6}$ fraction

of extra deflection beyond 20 Mpc in the formal isolated projection characterize scale separation and tails. They are not complete physical error bounds. A time-dependent material/environmental continuation consumes the specific equation of state or support and constrained outer data derived in Supplementary Note S5.

The leading scalar principal coefficients are positive on the retained branch (Appendix C); its maximum radial characteristic speed squared is approximately 1.474 in metric light-speed units. Photons in this benchmark follow metric null rays. This leading coefficient check establishes neither full coupled stability nor a causal ultraviolet completion. The independent rolling scalar T, target phase and metric coordinate retain their stated normalizations; the earlier source-free target null is preserved in Appendix A.

The BHSM microscopic source question has a precise boundary in this paper. Supplementary Note S4 derives the cap/matcher and Schur source pullback and gives positive same-domain complements with identical leading normalization but different higher-derivative coefficients. A physical cap/interface embedding into canonical T, its cubic vertex, source pairing and complement length/spectrum must therefore be supplied by the framework before a unique microscopic state is assigned to this transfer. This result supplies the defining conditions and their nonuniqueness; it does not select an extra optical amplitude. The present calculation is a reproducible effective benchmark against which that physical source construction can be tested. The original microscopic current remains

$$j_c = \frac{\mu}{2} \sum_s \Big[U_s^\dagger (iG_c) U_s - V_s^\dagger (iG_c) V_s \Big]. \tag{64}$$

The classical density and reduced J_{eff} evaluated here do not compute these sections or an achieved microscopic current. Their identification consumes the cap-to-canonical-field source pairing and same-domain state/Cauchy prescription; those are not fixed by this optical fit. The exclusive chronological two-point, inclusive contour four-point and complete matched-occupation constructions remain distinct. Neither a classical variability process nor inherited retarded arrays or a native heat functional supplies the statistical state kernel. No vacuum, dressed-state factorization or Wick replacement is selected, and the later four-point is not imposed as a prerequisite for the conditional classical optics used here.

12. Conclusions

For the prescribed $10^{11} M_\odot$, 3 kpc body at $z_l = 0.3$ and common source at $z_s = 1$, the fixed R1 action/state produces a regular source, screening radius approximately 212 kpc, reference-relative delay offsets 25.96 s and 8.65 s, image changes approximately 0.79–6.12 microarcsec and a 6.82 ppm delay-ratio change. The state dependence enters explicitly through the normalized scalar mixing and stiffness. These results retain one physical incident-light, clock and propagation prescription throughout.

The same retained metric and light transfer admit a positive conventional effective mass. A finite five-parameter ordinary refit also reproduces the three injected images and two delays to numerical precision; fixing its center leaves millisecond-scale delay mismatches in the declared profile family. The interpretation of the response therefore depends on the admitted lens freedoms. A smooth Plummer control matched by total mass and half-mass radius gives offsets 22.45 s and 9.62 s while preserving the effective-mass equivalence. The eight paired profile/input controls, common ordinary-GR correction and distributed optical encounter make the benchmark's model and approximation dependencies explicit. The common GR2 changes of the original offsets remain $+69$ and -10 microseconds.

The observed HE 0435 baseline trains on 902 epochs and reserves 100 later epochs. With later B as the allowed driver, conventional and R1 predictions for 300 A/C/D targets differ by only 0.00498 in joint log score. Their nominal 95% intervals cover 85.33% of these targets, and the separately retained all-channel forecast also undercovers. The first experiment exposes its brightness and central-image limitations. Actual HST F814W pixels constrain its unattenuated central/A ratio below 0.030 within the tested conditional PSF/noise family, versus the original 0.0632 prediction. A positive compact-source revision reduces that ratio to about 0.00264; the nearly tied allowed source candidates do not measure a nuclear mass. A separately committed test then evaluates the revised predictions on 162 unused LCO A/B/D magnitudes with C as the allowed driver, and all 216 magnitudes without a driver. Both revised macro models cover 160 of 162 primary targets, with approximately 0.0772 mag RMSE and 0.670 mag mean 95% interval width. Their nominal 68.27% intervals cover 94.44%, indicating conservative uncertainty. The primary R1-minus-conventional joint score is about -3.46×10^{-5}. These applications constrain source and predictive adequacy without identifying a unique BHSM envelope state or establishing sharp nominal probability calibration. The compact-source model's separate unused Gaia test leaves a 3.26 mas coordinate RMS and rejects the exact frozen point predictions under catalogue errors alone. Sensitivity to prespecified uncertainty floors does not establish precision validity. Diagnosing the old geometry motivates the explicit positive rear G1 and separately specified quadratic remainder. Full ordered two-plane propagation reduces consumed-Gaia RMS to about 0.011 mas without changing catalogue errors. Its revised central/A prediction is about 0.00165 and remains permitted by the consumed conditional HST test. These development fits establish neither a unique geometry nor an envelope-state measurement. The separately frozen JWST experiment in Section 9.7 improves raw geometry RMS from 3.31 to 0.779 mas on previously unused photons, but fails its instrumental and exposure-consistency gates. Neither conventional nor R1 receives a validation pass. The remaining precision-inference task is sharpened by an explicit valid-rate correction and two new consumed-data applications. Restoring 303 valid slopes gives 0.448 mas raw RMS while all profiles still fail. An electron-conserving history term does not repair the held optical mean. A coherent sky with shared calibration uncertainty lowers training residuals at a provisional coupled iterate, but also fails a held-exposure profile. Precision inference therefore requires an adequate instrumental mean and stochastic/reference covariance, with the physical-state and global-matching dependencies still explicit.

Prepared light constrains an envelope encounter transfer. Identifying that transfer with a particular underlying state consumes the specified action/source map and independently justified mass/environment data. The calculation provides a coherent forward and inverse test problem for that inference, with its physical source conditions and technical continuation derived separately.

Supplementary Materials: The repository supplies Notes S1–S10: local action/source reduction; finite-body GR2; nuisance optics; cap/interface source pullback; exact field/environmental continuation; profile robustness; finite nonlinear refit; distributed-encounter accuracy control; observed-lens source/two-plane derivation; and classical light-curve likelihood and reserved-target protocol. Their source paths, executable commands, input hashes, numerical outputs and focused checks are mapped in the acceptance checkpoint. The long cap and continuation TeX derivations are retained as technical supplementary source; the historical target and transport calculations remain available.

Author Contributions: Norman P. Carberry: conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation, writing—review and editing, and visualization. The author accepts responsibility for the scientific content. AI assistance is disclosed below.

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Informed Consent Statement: Not applicable.

Data Availability Statement: The numerical code, frozen inputs, outputs, source applications and derivation checks are archived in the public repository at calculation revision `64046bf8e15c` in the Manuscript-Generation calculation archive. The frozen background and prior controls are preserved from revision `49a1dab`. The accompanying manuscript sources, official journal dependencies and integration reviews are supplied with this manuscript. The observed photometry comprises the public COSMOGRAIL and LCO/CDS releases cited above. Exact HST science/weight/context cutouts and original MAST product URLs are also retained. Original files, attribution, hashes and reduction caveats accompany the data. The Gaia DR3 query, raw catalogue rows and separate pre-query prediction freeze are preserved as well. The later G1 geometry, field calibrations and first JWST application are preserved at immutable revision `7ad3cf175666`. This includes the separate pretarget protocols, independent LMC PSF file, exact native target windows, embedded official GWCS exports, raw residuals and failed adequacy diagnostics. Full native JWST files remain available from MAST under the recorded product URLs and hashes; the complete 470 MB native streams are not duplicated in the repository. The additive native-rate repair, byte-preserved supplied package, actual read-group and coherent-sky applications, residual maps and independent computational reviews are archived at `34694083c22d`. The supplied detector processing is preserved without a new replay; the new applications and their failed adequacy gates are separately identified. Reserved observations are not included in the development fits.

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Appendix A. The Constant-Target Null Control

The earlier source-free target sector has

$$S_\eta = -\int N\sqrt{h}\, w_0 F(X)\, \mathrm{d}t\, \mathrm{d}^3x, \qquad F(X) = \frac{\kappa_\eta}{2}X + \frac{1}{8}X^4, \qquad X = g^{\mu\nu}\langle D_\mu\eta, D_\nu\eta\rangle, \quad \text{(A1)}$$

with $\eta = (s, z) \in S^6 \subset \mathbb{R}\oplus\mathbb{C}^3$, $s^2 + 2z^\dagger z = 1$, target metric $\mathrm{d}s^2 + 2\,\mathrm{Re}(\mathrm{d}z^\dagger \mathrm{d}z)$, and $w_0\kappa_\eta > 0$. For $\eta_q = (\cos q, \sin q\, e_1/\sqrt{2})$ with constant q and zero connection, $D\eta = 0$, $F(0) = 0$, and the target current and stress vanish. For a radial meridian $q(r)$, regularity sets the conserved flux

$$r^2 e^\nu \sqrt{f_r}\, w_0(\kappa_\eta + X^3)q' = 0, \qquad X = f_r(q')^2 \quad \text{(A2)}$$

to zero. Positivity gives $q' = 0$. This proves the null only for the stated source-free regular boundary problem.

A constant background still permits positive perturbation stiffness. For a nonconstant Dirichlet phase perturbation, the quadratic energy and its second variation (phase self-curvature) are

$$E_\eta^{(2)}[\varphi] = \tfrac{1}{2} w_0 \kappa_\eta \int e^{\Phi-\Psi} |\nabla\varphi|^2 \, \mathrm{d}^3x,$$

$$\delta^2 E_\eta[\varphi,\varphi] = 2E_\eta^{(2)}[\varphi] > 0.$$

The mixed metric/phase response vanishes at $D\eta = 0$. The solved matter body therefore has ordinary nonzero lensing and zero additional response in this target family. A vanishing matched seam contribution is not a statement that all possible gravitational or boundary sources vanish. The rolling scalar T in Equation (4) is a different explicitly specified field.

Appendix B. Moving Measured-Redshift Endpoints

For an observer-normalized affine ray, $J_a'' = \mathcal{T} J_a$ with $(J_a, J_a')_a = (0, \omega_a I)$. Let $Z_{\lambda,a} = \delta \ln\omega|_{\lambda_a}$ and $Z_{\chi,a} = \delta\ln\omega|_{\chi_a}$. On the reference ray,

$$Z_{\chi,a} = Z_{\lambda,a} - \mathcal{H}_a \delta\chi_{\mathrm{ray},a}|_\lambda,$$

$$\delta\lambda_a = -\frac{Z_{\lambda,a}}{(\ln\overline{\omega})'_{\lambda,a}}, \qquad \delta\chi_a = \delta\chi_{\mathrm{ray},a}|_\lambda + \overline{\chi}'_{\lambda,a}\delta\lambda_a = -\frac{Z_{\chi,a}}{\mathcal{H}_a}. \tag{A3}$$

Thus converting an affine shift alone does not give the full radial event shift. The moving-vertex initial conditions are

$$\delta J_a|_a = -\omega_a I\,\delta\lambda_a, \qquad \delta J_a'|_a = \Delta\omega_a I, \qquad \Delta\omega_a = \delta\omega_a|_\lambda + \overline{\omega}'_a\delta\lambda_a. \tag{A4}$$

At fixed measured redshift and observer normalization, $\Delta\omega_a = 0$. Output event and frame conversions must also be included. These conventions are needed for the three compatible maps; they are inactive in the local comparison where the maps are held fixed.

Appendix C. Leading Scalar Principal Coefficients

After eliminating the leading metric mixing, the homogeneous time coefficient is

$$D = 1 + \frac{12Hv}{\lambda} + \frac{6v^4}{M_{\mathrm{Pl}}^2\lambda^2} = 4.3962210287. \tag{A5}$$

For $\mathcal{S} = \sqrt{C^2 + 16 b_T GM(<r)/(\lambda r^3)}$, the leading local diagonal coefficients are

$$K_r = \mathcal{S}, \qquad K_\Omega = \mathcal{S} + \frac{1}{2}r\mathcal{S}', \qquad K_t = D + \frac{3}{2}(\mathcal{S} - C) + \frac{1}{2}r\mathcal{S}'. \tag{A6}$$

Nonnegative density implies $K_\Omega \geq (\mathcal{S}^2 + 3C^2)/(4\mathcal{S}) > 0$. For this branch K_t is also positive; the frozen homogeneous ratio is $C/D = 0.9642051$. The exterior maximum of K_r/K_t is 1.47366 at approximately 100.107 kpc. These are leading local coefficients. Mixed time–space and higher weak-field terms, the full coupled spectrum, and an ultraviolet completion are outside this check.

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